

Metabolic regulation of misfolded protein import into mitochondria

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Abstract

Mitochondria are the cellular energy hub and central target of metabolic regulation. Mitochondria also facilitate proteostasis through pathways such as the ‘mitochondria as guardian in cytosol’ (MAGIC) whereby cytosolic misfolded proteins (MPs) are imported into and degraded inside mitochondria. In this study, a genome-wide screen in yeast uncovered that Snf1, the yeast AMP-activated protein kinase (AMPK), inhibits the import of MPs into mitochondria while promoting mitochondrial biogenesis under glucose starvation. We show that this inhibition requires a downstream transcription factor regulating mitochondrial gene expression and is likely to be conferred through substrate competition and mitochondrial import channel selectivity. We further show that Snf1/AMPK activation protects mitochondrial fitness in yeast and human cells under stress induced by MPs such as those associated with neurodegenerative diseases.

eLife assessment

This study makes an **important** connection between cellular metabolism and proteostasis through MAGIC, a previously proposed protein quality control pathway of clearance of cytosolic misfolded and aggregated proteins by importing into mitochondria. The authors reveal the role of Snf1, a yeast AMPK, in preventing the import of misfolded proteins to mitochondria for MAGIC controlled by the transcription factor Hap4, depending on the cellular metabolic status. The key message is **important**, although the evidence for physiological relevance of MAGIC for overall cellular proteostasis and its molecular regulation by Snf1 remains **incomplete**.

Introduction

Mitochondria are vital organelles whose biogenesis and activities in energy production are tightly linked to cellular metabolic control ([1](#) , [2](#) ). Metabolic stress and mitochondrial dysfunction are common drivers of age-related degenerative diseases such as heart failure and dementia

(3,4), which are often characterized by loss of proteostasis leading to the formation of protein aggregates (3,5). In yeast, acute proteotoxic stress such as heat shock induces reversible protein aggregation in cytosol (6-10). Protein aggregates are initially formed on the cytosolic surface of the endoplasmic reticulum (ER), and later captured at the mitochondrial outer membrane (7,8). Upon reversal to the stress-free condition, aggregates undergo dissolution that is not only dependent on the activity of the Hsp104 chaperone but also mitochondrial membrane potential (8,10). This observation led to a hypothesis that mitochondria play an active role in the clearance of cytosolic MPs. Using both imaging-based as well as biochemical assays, we showed that certain aggregation-prone native cytosolic proteins and the model aggregation protein firefly luciferase single mutant (FlucSM) (11), but not stable cytosolic proteins, are imported into the mitochondrial matrix (10). A subset of highly aggregation-prone proteins known as super-aggregators (9) are imported into mitochondria even in the absence of heat stress (10). Mitochondrial proteases, most prominently the LON protease Pim1, degrade the imported MPs in the mitochondrial matrix, and this pathway of clearance of cytosolic MPs was termed 'mitochondria as guardian in cytosol' (MAGIC) (10) (**Figure 1—figure supplement 1A**).

Cytosolic MPs have also been found in human mitochondria. Both FlucSM and a more destabilized double mutant (FlucDM) (11), but not the well-folded wild-type Fluc (FlucWT) or glutathione S-transferase (GST), are imported into the mitochondrial matrix of human RPE-1 cells (10). In HeLa cells, proteasomal inhibition by MG132 induces the mitochondrial import of unfolded cytosolic model protein in a manner dependent on mitochondrial outer membrane protein FUNDC1 and cytosolic chaperone HSC70 (12). Furthermore, disease-related proteins such as α -synuclein (α Syn), FUS, and TDP-43 are found in the mitochondria of human cells (13-15). These results suggest that a MAGIC-like pathway may exist in higher organisms, although the underlying mechanisms could be different.

It remains unclear whether MAGIC is beneficial or detrimental to cellular or mitochondrial fitness. Nevertheless, the MAGIC pathway may represent a link between mitochondrial dysfunction and loss of proteostasis. Although inhibition of mitochondrial import after heat shock causes prolonged protein aggregation in cytosol, an elevated burden of MPs in mitochondria can also cause mitochondrial damage (16). Understanding how mitochondria balance functions in proteostasis and metabolism may provide key insights into the maintenance of cellular fitness under stress during aging. In this work, we conducted an unbiased imaging-based genetic screen in yeast to uncover cellular mechanisms that regulate MAGIC. We identified Snf1, the yeast AMP-activated kinase (AMPK), as a negative regulator of MAGIC through transcriptional upregulation of nuclear-encoded mitochondrial genes. We also showed that AMPK activation in yeast and human cells attenuates mitochondrial accumulation of disease-related MPs and may protect cellular fitness under proteotoxic stresses.

Results

A genetic screening for regulators of MAGIC

To observe the mitochondrial import of cytosolic MPs, we employed a previously established method using split-GFP (spGFP) system in which the first ten β -strand of GFP (GFP₁₋₁₀) was targeted into the mitochondrial matrix while the eleventh β -strand (GFP₁₁) was tagged with MPs (10) (**Figure 1—figure supplement 1B**). Because mitochondrial import requires substrate in an unfolded state (17), globular GFP reconstituted in the cytosol should not be imported. Indeed, mitochondrial spGFP signal of stable cytosolic protein Hsp104 failed to increase after heat shock (10) (**Figure 1—figure supplement 1C**). In contrast, spGFP signals of FlucSM and several endogenous aggregation-prone proteins increased significantly after heat shock at 42 °C compared to background at normal growth temperature (30 °C) in wild-type cells (10). Importantly,

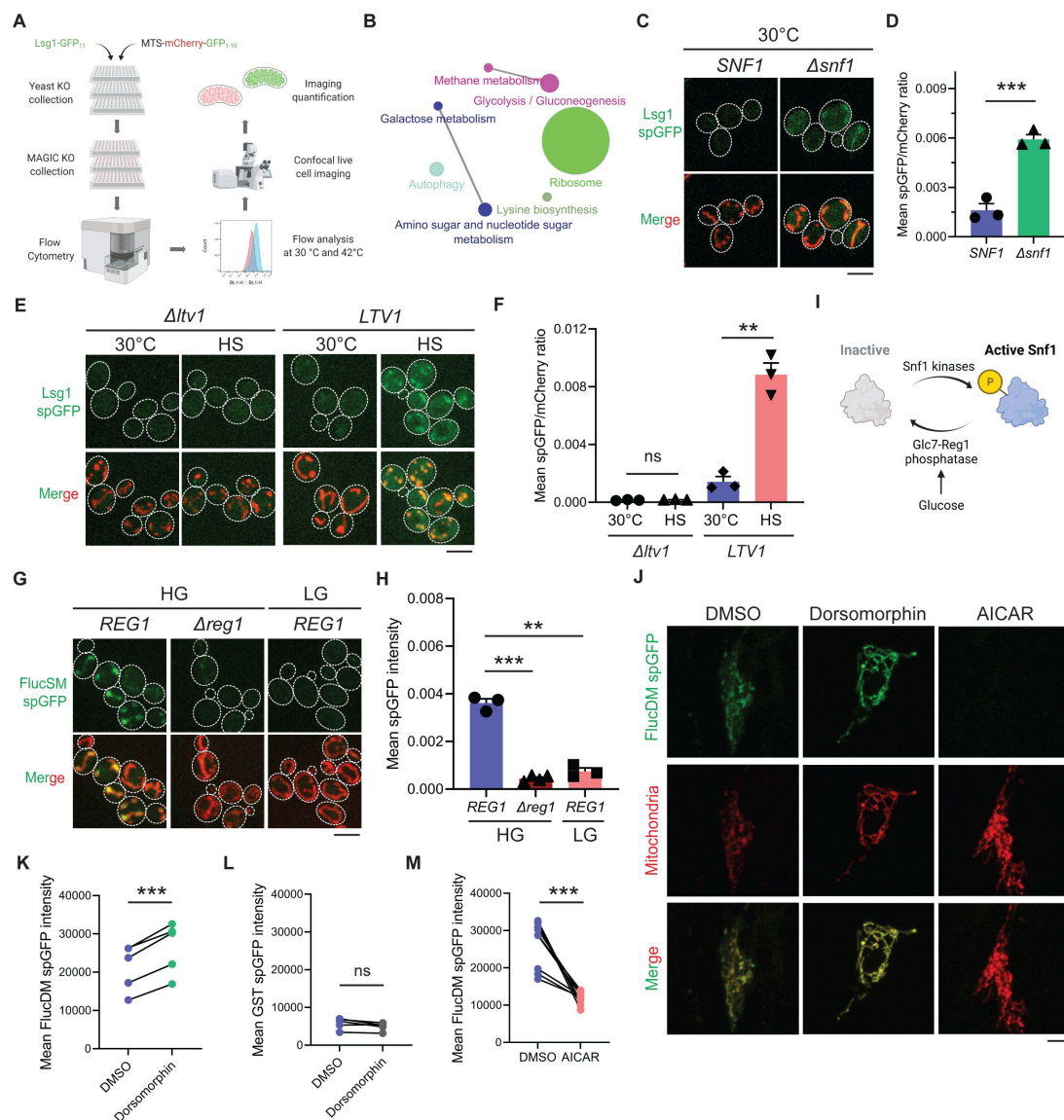


Figure 1.

MAGIC regulators revealed by a genome-wide screen in yeast and validations in human RPE-1 cells.

(A) Workflow of the spGFP-based genetic screen in yeast. (B) KEGG pathway analysis of validated mutants that affect MAGIC. The size of the node indicates the number of genes identified. Pathways with at least two associated genes are shown. (C, D) Representative images (C) and quantification (D) of Lsg1 spGFP signal in WT and $\Delta snf1$ cells at 30 °C. Shown in (C): Top, Lsg1 spGFP; Bottom, merged images of spGFP and mitochondria labeled with MTS-mCherry. Shown in (D): means $\pm$ SEM of spGFP/mCherry ratio. Unpaired two-tailed *t*-test. (E, F) Representative images (E) and quantification (F) of Lsg1 spGFP signal in $\Delta ltv1$ and wild-type *LTV1* cells at 30 °C and after HS. Shown in (F): means $\pm$ SEM of spGFP/mCherry ratio. Paired two-tailed *t*-test. HS: heat shock. (G, H) Representative images (G) and quantification (H) of FlucSM spGFP signals in WT (*REG1*) cells in HG or LG, and $\Delta reg1$ cells in HG. Shown in (G): Top, FlucSM spGFP; Bottom, merged images of spGFP and mitochondria labeled with Tom70-mCherry. Shown in (H): means $\pm$ SEM of spGFP intensity. Paired (*REG1* in HG vs. LG) or unpaired (*REG1* vs. $\Delta reg1$ in HG) two-tailed *t*-test. (I) Schematic diagram of Snf1 activation in yeast. (J) Representative images of FlucDM spGFP in RPE-1 cells treated with DMSO, dorsomorphin, or AICAR. Top, FlucDM spGFP; Middle, mitochondria-targeted mCherry; Bottom, merged images. (K–M) Flow cytometry-based quantifications of FlucDM spGFP in RPE-1 cells treated with DMSO, dorsomorphin, or AICAR (K, M), and GST spGFP in cells treated with DMSO or dorsomorphin (L). Means $\pm$ SEM of spGFP intensities are shown. Paired two-tailed *t*-test. ***P* < 0.01; ****P* < 0.001; ns, not significant, *P* > 0.05. HG: 2% glucose; LG: 0.1% glucose plus 3% glycerol. Scale bars, 5 μ m.

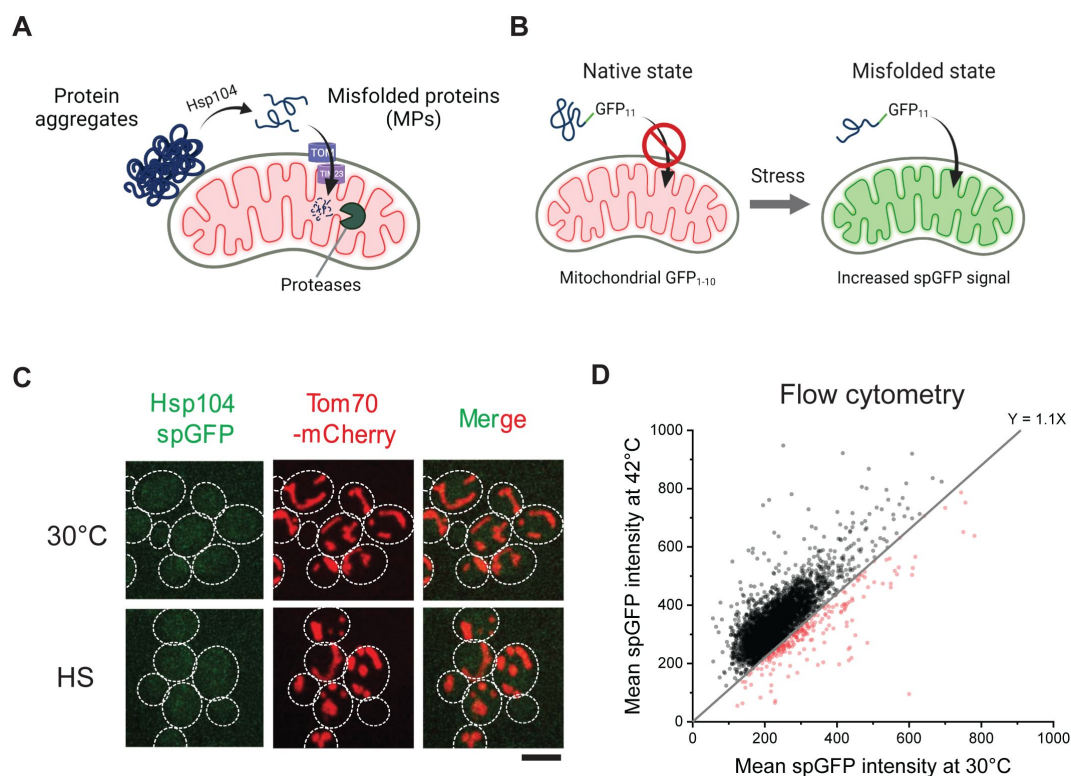


Figure 1—figure supplement 1.

Schematics of MAGIC pathway and spGFP-based imaging in the whole-genome screen in yeast.

(**A**) Schematic diagram of MAGIC involving the import of cytosolic misfolded proteins through mitochondrial import machineries and the subsequent degradation under proteotoxic stresses (heat shock or overload of MPs). (**B**) Schematic diagram of spGFP reporter. (**C**) Hsp104 spGFP in mitochondria at 30 °C and after heat shock (HS) at 42 °C for 30 min. Endogenous Hsp104 was tagged with GFP₁₁, while GFP₁₋₁₀ was constitutively targeted to the mitochondrial matrix by linking to a matrix protein Grx5. Similar results were obtained by using MTS-mCherry-GFP₁₋₁₀ (100000). 3 biological repeats, 63 cells (30 °C) and 220 cells (HS) imaged. Scale bar, 5 μm. (**D**) Flow cytometry readouts of yeast knockout mutants at 30 °C and after 42 °C HS. Red dots represent mutants that failed to show an increase in Lsg1 spGFP after HS.

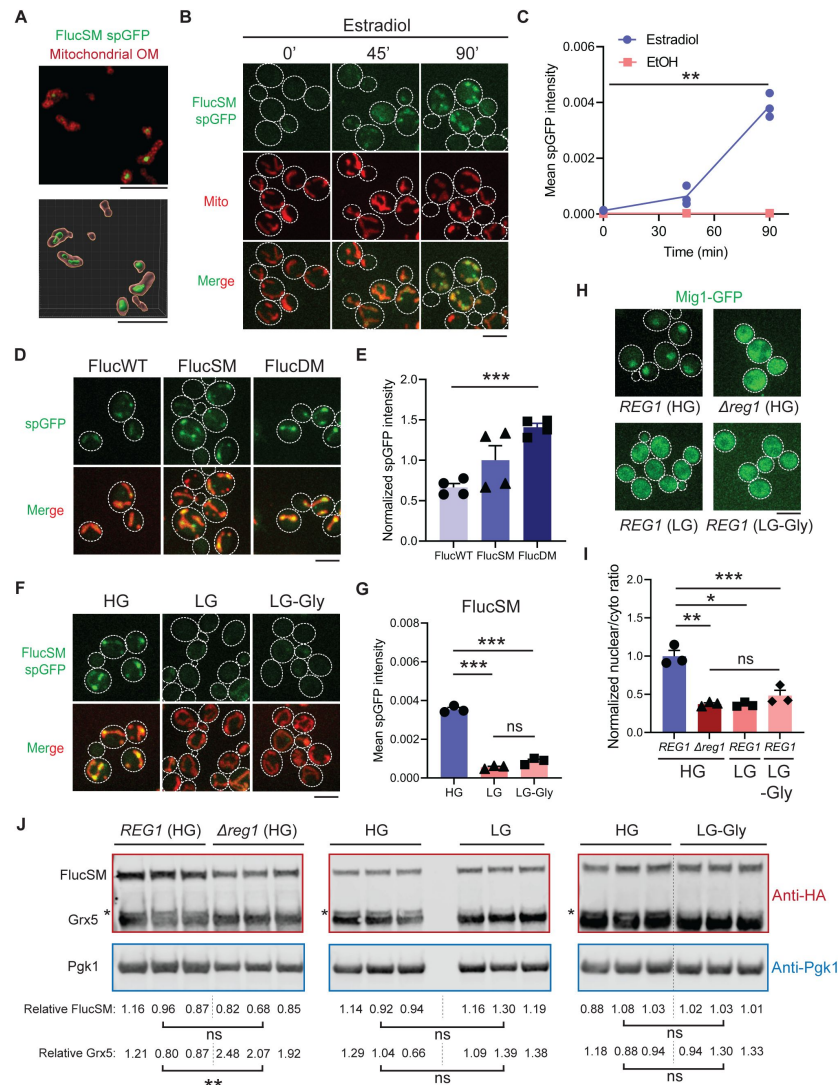


Figure 1—figure supplement 2.

Snf1 regulates the accumulation of misfolded proteins in mitochondria after acute overexpression of FlucSM.

(A) Representative super-resolution imaging for FlucSM spGFP signal in mitochondria after 90 min estradiol induction. Top, maximum projection. Bottom, 3D rendering. OM: outer membrane labeled by Tom70-mCherry. (B, C) Representative images (B) and quantification (C) of time-dependent accumulation of FlucSM spGFP signal in mitochondria at 30 °C after estradiol or ethanol treatment in HG medium. Shown in (B): Top, FlucSM spGFP; Middle: mitochondria labeled with Tom70- mCherry; Bottom: merged images. Shown in (C): means $\pm$ SEM of spGFP intensity. Paired two-tailed *t*-test comparing 0 min and 90 min estradiol treatment. (D, E) Representative images (D) and quantification (E) of FlucWT, FlucSM, and FlucDM spGFP after 90 min estradiol treatment in HG medium. Shown in (E): means $\pm$ SEM of normalized spGFP intensities. Unpaired two-tailed *t*-test between FlucWT and FlucDM. (F, G) Representative images (F) and quantification (G) of FlucSM spGFP in cells that grew in HG, LG, and LG-Gly media. Shown in (G): means $\pm$ SEM of spGFP intensity. Paired two-tailed *t*-test. (H, I) Representative images (H) and quantification (I) of the nuclear-cytoplasmic translocation of Mig1-GFP. Shown in (I): means $\pm$ SEM of normalized Mig1-GFP nuclear-cytoplasmic ratio. Paired (WT in HG vs. LG or LG-Gly) or unpaired (WT vs. $\Delta reg1$ in HG) two-tailed *t*-test, and one-way ANOVA for comparing $\Delta reg1$ in HG, WT in LG, and WT in LG-Gly). (J) Immunoblots of FlucSM-HA-GFP₁₁ and Grx5-HA-GFP₁₋₁₀ in cell lysates. Relative FlucSM levels (FlucSM/Pgk1) and relative Grx5 levels (matured Grx5/Pgk1) of three biological repeats are shown. Paired (HG vs. LG or LG-Gly) or unpaired (WT vs. $\Delta reg1$ in HG) two-tailed *t*-test. Asterisks indicate the precursor species of Grx5-HA-GFP₁₋₁₀ that are excluded from quantification. HG: 2% glucose; LG: 0.1% glucose plus 3% glycerol; LG-Gly: 0.1% glucose. EtOH: ethanol. **P* < 0.05; ***P* < 0.01; ****P* < 0.001; ns, not significant, *P* > 0.05. Scale bars, 5 μ m.

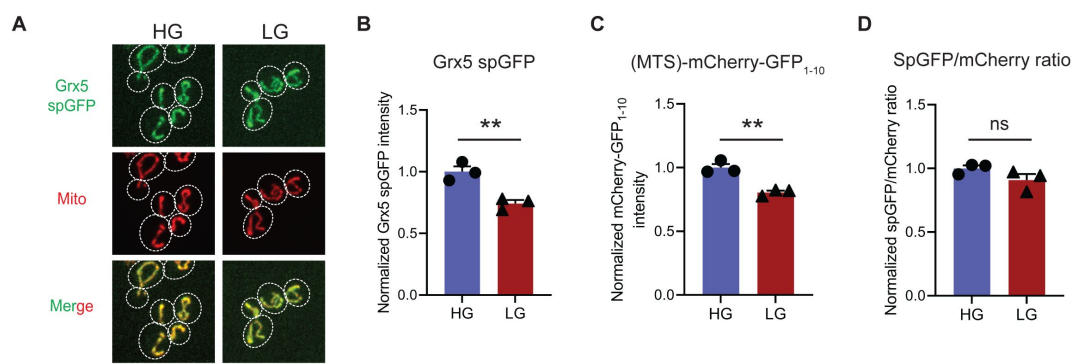


Figure 1—figure supplement 3.

Snf1 activation only modestly affects spGFP reconstitution.

(A-D) Representative images (A) and quantification of Grx5 spGFP (B), MTS-mCherry-GFP1-10 (C) and spGFP-to-mCherry ratio (D) in HG or LG medium. Means $\pm$ SEM are shown in (B-D). Unpaired two-tailed *t*-test. ***P* < 0.01; ns, not significant, *P* > 0.05. HG: 2% glucose; LG: 0.1% glucose plus 3% glycerol. Scale bars, 5 μ m.

mitochondrial import of FlucSM and other misfolded cytosolic proteins after heat stress was further validated by using a variety of additional methods, including the classical biochemical fractionation and protease protection assay, APEX-based labeling in mitochondrial matrix, and super-resolution microscopy (10).

To uncover cellular pathways that influence MAGIC, we performed a high-throughput spGFP-based genetic screen in the non-essential yeast knockout (YKO) collection (18) (Figure 1A). Briefly, for each mutant strain in this collection, Lsg1, one of the endogenous aggregation-prone proteins previously shown to be imported into mitochondria (10), was C-terminally tagged with GFP₁₁ at *LSG1* genomic locus through homologous recombination. Also introduced into each mutant strain was a construct constitutively expressing matrix targeted GFP₁₋₁₀ under the GAPDH promoter. GFP₁₋₁₀ was targeted into mitochondrial matrix by using the cleavable mitochondrial targeting sequence (MTS) of Subunit 9 of mitochondrial ATPase (Su9) from *Neurospora crassa*, and the red fluorescent protein mCherry was also included in this construct (MTS-mCherry-GFP₁₋₁₀), as previously described (10). YKO mutants bearing the above Lsg1 spGFP reporter components were generated by using high-throughput transformation in 96-well plates. We used flow cytometry and analyzed Lsg1 spGFP signal of each mutant at 30 °C and after 42 °C heat shock for 30 min (Figure 1—figure supplement 1D). Mutants of interest were then subjected to hits validation using confocal fluorescence imaging. Based on mitochondrial spGFP intensity of each mutant and wild-type (WT) cells at two imaging time points, we classified the validated YKO mutants into two groups: five Class 1 mutants showed significant greater spGFP signal than WT at 30 °C without heat shock, and 140 Class 2 mutants had no significant increase in spGFP signal after heat stress compared to 30 °C (Table S1; Details in Materials and Methods).

KEGG pathway analysis revealed that genes corresponding to the hits validated with imaging encompassed many cellular pathways, most notably carbohydrate metabolism and ribosomal biogenesis (Figure 1B). Among five Class 1 mutants, a notable one is *Snf1* (Figure 1C and D; See further analyses below). Class 2 includes multiple genes related to ribosomal biogenesis (Table S1). For example, deletion of *LTV1* that encodes a chaperone required for the assembly of small ribosomal subunits (19) showed only baseline level Lsg1 spGFP fluorescence with no increase at 42 °C (Figure 1E and F).

Snf1/AMPK negatively regulates MP accumulation in mitochondria

In this study, we have chosen to focus on *SNF1*, as *SNF1* encodes the yeast homolog of the evolutionarily conserved AMPK which serves as a master nutrient sensor orchestrating the activation of glucose-repressed gene transcription and metabolic stress response in glucose-limited conditions (20–22). Its pivotal function in cellular metabolism and mitochondrial biogenesis spurred us to further examine its role in MAGIC. To avoid complicating effects of heat shock and to improve the sensitivity of spGFP reporter, we optimized our spGFP-based method to impose proteostasis burden by acute induction of the MAGIC substrate FlucSM (10,11) tagged with GFP₁₁ (FlucSM-GFP₁₁) via the β -estradiol-inducible system (23). GFP₁₋₁₀ was stably targeted to the mitochondrial matrix by fusion with a matrix protein Grx5 (Grx5-GFP₁₋₁₀). After induction upon β -estradiol treatment at 30 °C for 90 min, FlucSM spGFP signal increased significantly within mitochondria compared to the ethanol-treated control (Figure 1G and H; Figure 1—figure supplement 2A–C; Movie S1). The spGFP signal in mitochondria showed an increasing trend that positively correlated with the structural instability of luciferase-derived MPs: FlucWT, FlucSM, and FlucDM with the highest structural instability (11) (Figure 1—figure supplement 2D and E).

We chose to use the intermediate construct, FlucSM-GFP₁₁, for testing the effects of modulating Snf1 activity on mitochondrial import of MPs.

Reg1 is the regulatory subunit of Glc7-Reg1 protein phosphatase 1 complex that dephosphorylates Snf1 and promotes its inhibitory conformation (24–27). Either glucose limitation or loss of Reg1 in glucose-rich medium (HG; 2% glucose) result in constitutive activation of Snf1 and relief from glucose repression of transcription (24–28) (Figure 1I). We found that $\Delta reg1$ cells exhibited significantly less accumulation of FlucSM in mitochondria than WT cells, and likewise, wild-type cells that grew in low glucose medium (LG; 0.1% glucose plus 3% glycerol) showed significantly lower FlucSM spGFP compared to cells in HG (Figure 1G and H). The absence of glycerol in LG (LG-Gly) did not cause any noticeable difference to LG (Figure 1—figure supplement 2F and G). Snf1 activation under these conditions was validated by the nuclear export of Mig1, which depends on phosphorylation by active Snf1 (29–31) (Figure 1—figure supplement 2H and I). In addition, the abundance of FlucSM-GFP₁₁ induced by estradiol was not affected by Snf1 activation, and Grx5-GFP₁₋₁₀ level was unchanged in low glucose media and even elevated in $\Delta reg1$ cells - a trend opposite of the spGFP changes (Figure 1—figure supplement 2J). These data exclude the possibility that reduced expression of either protein led to lower spGFP signal in mitochondria. To examine the effect of Snf1 activation on spGFP reconstitution, Grx5 spGFP strain was constructed in which the endogenous mitochondrial matrix protein Grx5 was C-terminally tagged with GFP₁₁ at its genomic locus, and GFP₁₋₁₀ was targeted to mitochondria through cleavable Su9 MTS (MTS- mCherry-GFP₁₋₁₀) (10). Only modest reduction in Grx5 spGFP mean intensity was observed in LG compared to HG, and no significant difference after adjusting the GFP₁₋₁₀ abundance (spGFP/mCherry ratio) (Figure 1—figure supplement 3A–D). These data suggest that any effect on spGFP reconstitution is insufficient to explain the drastic reduction of MP accumulation in mitochondria under Snf1 activation. Overall, our results demonstrate that Snf1 activation primarily prevents mitochondrial accumulation of MPs, but not that of normal mitochondrial proteins.

We previously showed that the import of firefly luciferase mutants into mitochondria of human RPE-1 cells was positively correlated with protein instability (10, 11). Using the established spGFP reporter, we found that treatment of RPE-1 cells with dorsomorphin, a chemical inhibitor of AMPK (32), significantly increased mitochondrial accumulation of FlucDM (Figure 1J and K), but not GST, a well-folded protein control (Figure 1L). In contrast, pharmacological activation of AMPK via 5-aminoimidazole-4-carboxamide ribonucleoside (AICAR) (33), significantly reduced FlucDM accumulation in mitochondria (Figure 1J and M). These results suggest that AMPK in human cells regulates MP accumulation in mitochondria following a similar trend as in yeast, although the underlying mechanisms might differ between these organisms.

Mechanisms of MAGIC regulation by Snf1

The accumulation of MPs in mitochondria as observed using the spGFP reporter should depend on the relative rates of import versus degradation by mitochondrial proteases, most prominently Pim1 – the conserved Lon protease in yeast (10). Three possible factors could therefore contribute to the reduced mitochondrial accumulation of MPs under Snf1 activation: 1) enhanced intra-mitochondrial degradation, 2) reduced cytosolic misfolded protein (due to enhanced folding and/or other degradation pathways), and 3) blocked mitochondrial import (Figure 2A). To evaluate the first possibility, an antimorphic mutant *pim1*^{S974D} was used to block the degradation of imported FlucSM in the mitochondrial matrix (34). Indeed, in HG medium wild-type cells overexpressing *pim1*^{S974D} showed a significantly increased accumulation of FlucSM in mitochondria compared to cells overexpressing *PIM1* (Figure 2B and C). However, *pim1*^{S974D} overexpression was unable to increase FlucSM accumulation in mitochondria of $\Delta reg1$ cells or WT cells growing in LG medium (Figure 2B and C). This result argued against the first possibility, and consistently the abundance of Pim1 protein was not increased by switching to nonfermentable carbon sources (35). To evaluate the second possibility, we used an *in vivo* firefly luciferase assay (36) and assessed the folding of enzymatically active FlucSM after estradiol induction. The result showed that Snf1-active cells exhibited reduced FlucSM luciferase activity, suggesting an increased rather than decreased fraction of misfolded FlucSM (Figure 2D). Furthermore, blocking the activated autophagy pathway in LG medium (37) did not

increase FlucSM spGFP in mitochondria (**Figure 2—figure supplement 1A** and **B**). We also observed that proteasomal inhibition through MG132 treatment stimulated the mitochondrial accumulation of FlucSM but did not ablate the difference between HG and LG condition (**Figure 2—figure supplement 1C**). The stimulating effect of MG132 was not surprising because FlucSM is degraded by proteasome in the cytosol (10) and preventing this pathway could divert more of such protein molecules toward MAGIC. We thus favor the third possibility that Snf1 activation specifically prevents the import of MPs into mitochondria.

Next, we investigated downstream transcription factors that could mediate the Snf1-regulated MP import (**Figure 2E**). In the presence of abundant glucose and when Snf1 activity is low, transcriptional repressor Mig1 and its partially redundant homolog Mig2 are localized in the nucleus to confer glucose-repressed gene expression (29–30, 38). However, neither single deletion of *MIG1* nor double deletions of *MIG1* and *MIG2* reduced FlucSM spGFP in HG medium (**Figure 2—figure supplement 1D** and **E**), suggesting that Mig1 and/or Mig2-repressed gene expression was not sufficient to prevent MP import (**Figure 2E**, left branch). Then we tested if MP import was antagonized by transcriptional activators downstream of Snf1 including Cat8, Hap4, Sip4, Atr1 and Rds2 (21, 39–41) (**Figure 2E**, right branch). Interestingly, only deletion of *HAP4*, but not other transcriptional activators, significantly rescued FlucSM import defect in $\Delta reg1$ cells with Snf1 activation (**Figure 2F and G**; **Figure 2—figure supplement 1F** and **G**). When cultured in LG medium, *HAP4* deletion also resulted in a significant increase in mitochondrial accumulation of FlucSM in comparison to WT (**Figure 2—figure supplement 1H**). Furthermore, overexpression of Hap4 alone was sufficient to reduce FlucSM spGFP in HG medium (**Figure 2H and I**). These data suggest that Hap4 is a main downstream effector of Snf1 that regulates MP import.

Hap4 is the transcriptional activation subunit in the Hap2/3/4/5 complex that activates the expression of nuclear encoded mitochondrial proteins and contributes to mitochondrial biogenesis during metabolic shifts or cellular aging (39–43). We hypothesized that elevated expression of mitochondrial preprotein induced by activation of Snf1-Hap4 axis (20, 35, 43–46) may outcompete MPs for import channels, especially considering that the expression of TOM complex components on the mitochondrial outer membrane was static in Snf1-active cells (35, 46) (**Figure 3A**).

To test this hypothesis, we attempted to restore MP import during Snf1 activation by using high-level expression of the soluble cytosolic domain of import receptors. The cytosolic import receptors lacking membrane-anchoring sequences are known to prevent mitochondrial preproteins from binding TOM complexes and thus inhibit preprotein import (47–49) (**Figure 3—figure supplement 1A**). Interestingly, overexpression of the cytosolic domain of Tom70 (Tom70_{cd}), but not Tom20_{cd} or Tom22_{cd}, significantly increased FlucSM import in LG medium (**Figure 3B and C**). Tom70_{cd} also further increased FlucSM import in HG medium (**Figure 3—figure supplement 1B** and **C**). The effect of Tom70_{cd} in cytosol required both the substrate binding and the chaperone-interaction domain (**Figure 3C**; **Figure 3—figure supplement 1D** and **E**). These results suggest that Tom70-dependent preprotein import may compete with misfolded protein import for limited TOM complexes. To further test if endogenous full-length Tom70 on the mitochondrial outer membrane is dispensable for MP import, we deleted *TOM70* and its paralog *TOM71* and found that in HG medium where mitochondrial respiration is not essential, FlucSM accumulation in mitochondria was not impaired in single mutants and increased in double mutant (**Figure 3D and E**). This result indicates that MP import does not use Tom70/Tom71 as obligatory receptors. The effect of $\Delta tom70\Delta tom71$ on MP import was consistent, albeit less pronounced, with Tom70_{cd} overexpression (**Figure 3D and E**; **Figure 3—figure supplement 1B** and **C**). One potential explanation for the modest effect in double mutant is that given to the functional redundancy between Tom20 and Tom70 (50, 51), Tom20 receptors in $\Delta tom70\Delta tom71$ cells could instead mediate preprotein import, whereas cytosolic Tom70_{cd} may have a dominant inhibitory effect on preprotein import by reducing association between

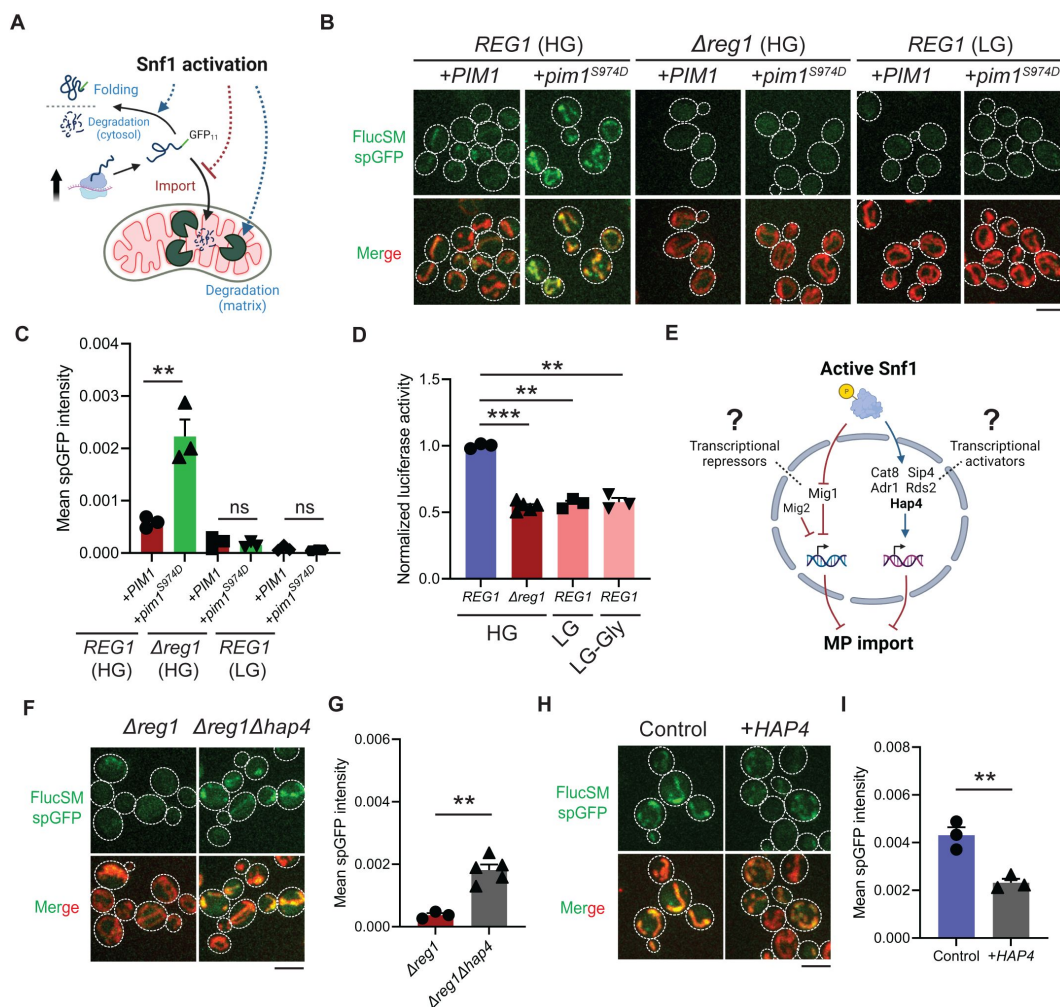


Figure 2.

Snf1 negatively regulates mitochondrial import of cytosolic misfolded proteins.

(A) Schematic diagram showing three possible explanations for reduced spGFP in mitochondria of Snf1-active cells: reduced MPs, blocked import, or enhanced degradation. (B, C) Representative images (B) and quantification (C) of FlucSM spGFP in Snf1-inactive and Snf1-active cells overexpressing copper-inducible *PIM1* or *pim1*^{S974D}. Shown in (C): means ± SEM of spGFP intensities. Unpaired two-tailed *t*-test. (D) Relative *in vivo* luciferase activity after 90 min of estradiol treatment. Means ± SEM of normalized FlucSM activity are shown. Paired (WT in different media) or unpaired (WT vs. *Δreg1* in HG) two-tailed *t*-test. LG-Gly: 0.1% glucose only. (E) Hypothetical regulations of import of MPs through transcriptional repressors and activators downstream of Snf1 activation. (F, G) Representative images (F) and quantification (G) of FlucSM spGFP in *Δreg1* and *Δreg1Δhap4* cells in HG medium. Shown in (G): means ± SEM of spGFP intensity. Unpaired two-tailed *t*-test. (H, I) Representative images (H) and quantification (I) of FlucSM spGFP in wild-type cells (control) or with constitutive overexpression of *HAP4* in HG medium. Shown in (I): means ± SEM of spGFP intensities. Unpaired two-tailed *t*-test. ***P* < 0.01; ****P* < 0.001; ns, not significant, *P* > 0.05. Scale bars, 5 μm.

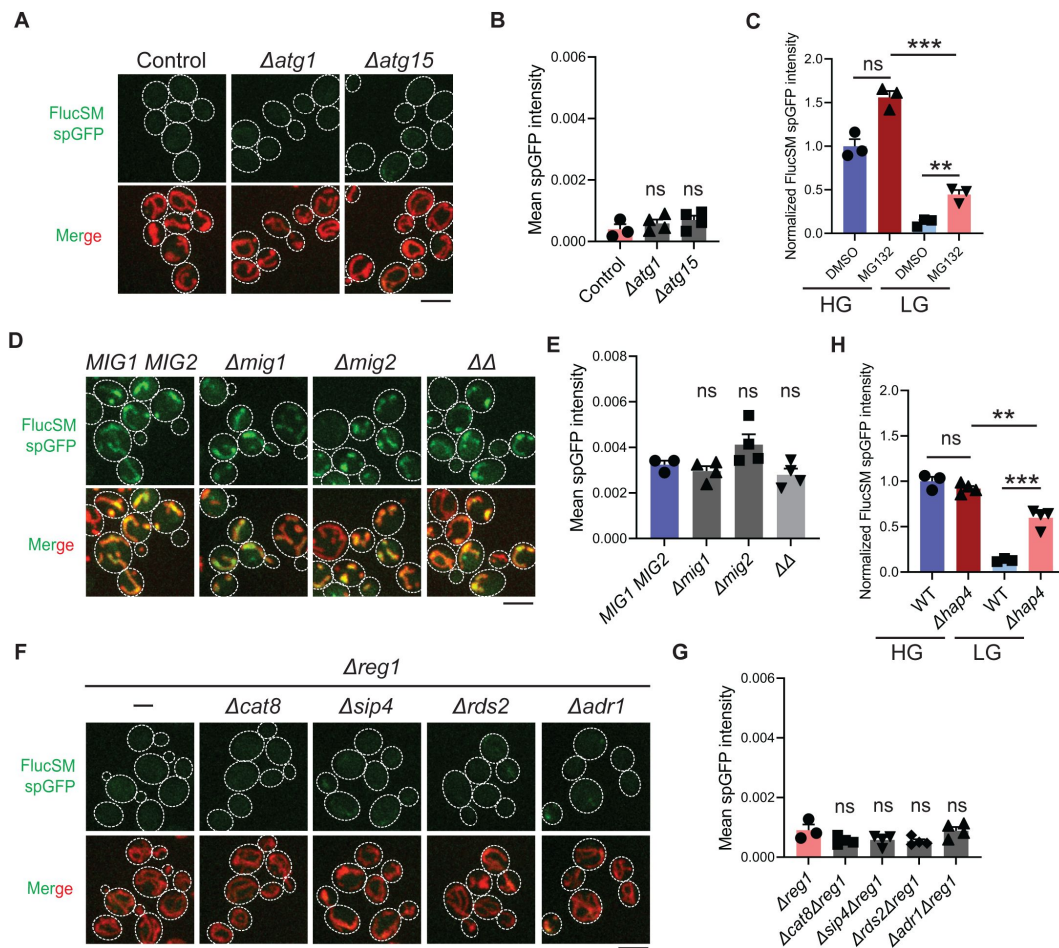


Figure 2—figure supplement 1.

Reduced accumulation of misfolded proteins in mitochondria under Snf1 activation is neither caused by elevated autophagy nor mediated by certain transcription factors.

(**A**, **B**) Representative images (**A**) and quantification (**B**) of FlucSM spGFP in control cells and autophagy-deficient mutants in LG medium. Shown in (**A**): means $\pm$ SEM of spGFP intensity. Unpaired two-tailed *t*-test between control and each mutant. (**C**) Quantification of FlucSM spGFP signals in $\Delta prd5$ cells treated with DMSO or 80 μM MG132 for 90 min during estradiol induction. Means $\pm$ SEM of normalized spGFP intensities are shown. Paired two-tailed *t*-test. (**D**, **E**) Representative images (**D**) and quantification (**E**) of FlucSM spGFP in wild-type ($MIG1 MIG2$), $\Delta mig1$, $\Delta mig2$, and $\Delta mig1 \Delta mig2$ ($\Delta \Delta$) cells in HG medium. Shown in (**E**): means $\pm$ SEM of spGFP intensity. Unpaired two-tailed *t*-test between WT and each mutant. (**F**, **G**) Representative images (**F**) and quantification (**G**) of FlucSM spGFP in $\Delta reg1$ and double mutant cells in HG medium. Shown in (**G**): means $\pm$ SEM of spGFP intensity. Unpaired two-tailed *t*-test between $\Delta reg1$ and each double mutant. (**H**) Quantification of FlucSM spGFP signals in WT or $\Delta hap4$ cells. Unpaired two-tailed *t*-test between WT and $\Delta hap4$. Pair two-tailed *t*-test for $\Delta hap4$ cells in different medium. ** $P < 0.01$; *** $P < 0.001$; ns, not significant, $P > 0.05$. Scale bars, 5 μm .

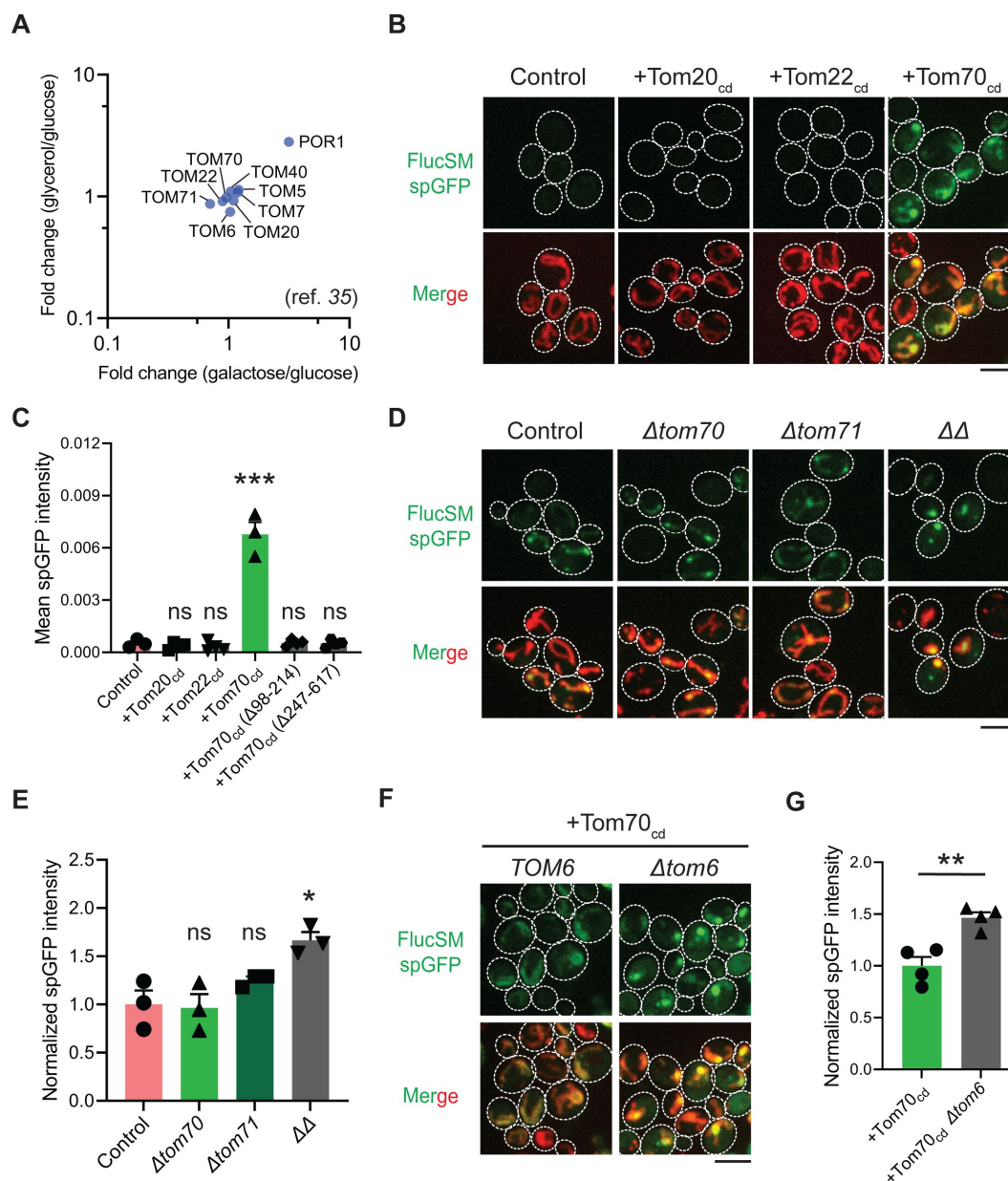


Figure 3.

Mechanisms underlying Snf1-regulated MP import into mitochondria.

(A) Fold changes in protein abundance of TOM complex components in glucose-limiting condition (glycerol or galactose) compared to glucose-rich condition. Raw data are retrieved from a published quantitative mass spectrometry dataset (35). (B, C) Representative images (B) and quantification (C) of FlucSM spGFP in wild-type control cells and cells overexpressing Tom20_{cd}, Tom22_{cd}, and Tom70_{cd} (Figure 3—figure supplement 1D) in LG medium. Shown in (C): means ± SEM of spGFP intensities. Unpaired two-tailed *t*-test between control and overexpression strains. (D, E) Representative images (D) and quantification (E) of FlucSM spGFP in wild-type control, Δtom70, Δtom71, and Δtom70 Δtom71 (ΔΔ) cells in HG medium. Shown in (D): Top, FlucSM spGFP; Bottom, merged images of spGFP and mitochondria labeled with mCherry-Fis1TM. Shown in (E): means ± SEM of normalized spGFP intensity. Unpaired two-tailed *t*-test. (F, G) Representative images (F) and quantification (G) of FlucSM spGFP in control and Δtom6 cells overexpressing Tom70_{cd} in LG medium. Shown in (G): means ± SEM of normalized spGFP intensities. Unpaired two-tailed *t*-test. **P* < 0.05; ***P* < 0.01; ****P* < 0.001; ns, not significant, *P* > 0.05. Scale bars, 5 μm.

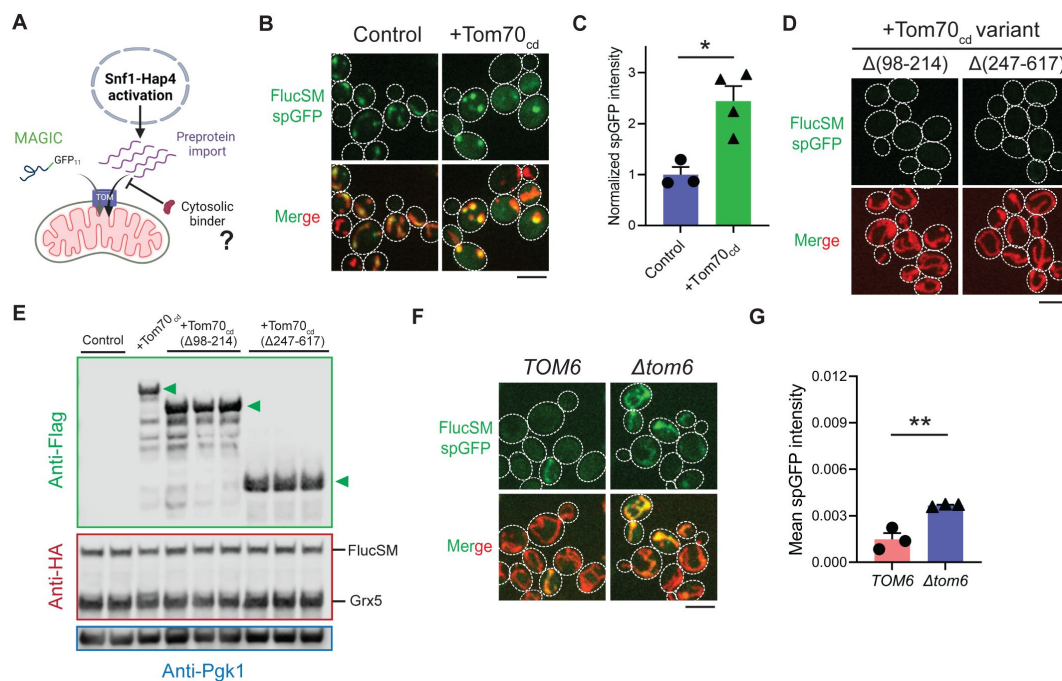


Figure 3—figure supplement 1.

Role of Tom70 cytosolic domain and Tom6 in regulating misfolded protein import.

(A) Working model of preprotein import and misfolded protein import through limited TOM channels. Introducing cytosolic binders of preproteins may release the import capacity for misfolded proteins. (B, C) Representative images (B) and quantification (C) of FlucSM spGFP in cells with or without Tom70_{cd} overexpression in HG medium. Shown in (C): means ± SEM of normalized spGFP intensity. Unpaired two-tailed *t*-test. (D) Representative images of FlucSM spGFP in cells overexpressing truncated Tom70_{cd} variants in LG medium. Quantification is shown in [Figure 4C](#). (E) Immunoblots of overexpressed Tom70_{cd}-3×Flag variants in lysates of cells that grew in LG medium. Anti-HA panel shows FlucSM-HA-GFP₁₁ after 90min estradiol treatment and constitutively expressed Grx5-HA-GFP₁₋₁₀. Arrowheads indicate intact Tom70_{cd}-3×Flag variants. (F, G) Representative images (F) and quantification (G) of FlucSM spGFP in WT and Δtom6 cells in LG medium. Shown in (G): means ± SEM of spGFP intensities. Unpaired two-tailed *t*-test. **P* < 0.05; ***P* < 0.01; ns, not significant, *P* > 0.05. Scale bars, 5 μm.

preproteins and mitochondrial outer membrane or TOM complexes (47–49). Together, these data suggest that increased expression and receptor-dependent import of certain mitochondrial preproteins under Snf1 activation might indirectly restrict the import of MPs.

As the main entry gate for mitochondrial preproteins, the TOM complex adopts two functional conformations with different substrate specificity: the receptor-free dimer is primarily responsible for importing MIA pathway substrates, and the receptor-bound trimer for Tim23 pathway substrates (52–54). Deletion of Tom6 disassembles the trimer and shifts the conformation equilibrium toward the dimer form (54, 55). To test if the substrate selectivity of TOM complex regulates MP import, we eliminated the trimer conformation by deleting *TOM6* and found that it elevated FlucSM import in LG medium with or without Tom70_{cd} overexpression (Figure 3F and G; Figure 3—figure supplement 1F and G). This result suggests that restricting MP import under Snf1 activation requires the trimeric TOM complex in addition to the competing mitochondrial preprotein import, and MPs might preferentially cross the mitochondrial outer membrane through the dimeric TOM complex.

AMPK protects cellular fitness during proteotoxic stress

We next investigated the physiological effects of metabolic regulation of MAGIC mediated by Snf1/AMPK. Prolonged induction of high-level FlucSM expression imposed a proteotoxic stress and led to a reduced growth rate in HG medium compared to the control, but interestingly no growth reduction was observed under glucose limitation (Figure 4A; Figure 4—figure supplement 1A, D and E). We reasoned that the lack of growth defect in LG medium could be due to prevention of MP import into mitochondria downstream of Snf1 activation. Supporting this, elevating MP import by Tom70_{cd} overexpression led to a reduced growth rate in LG medium that was dependent on FlucSM expression (Figure 4A; Figure 4—figure supplement 1B). Tom70_{cd} overexpression also exacerbated growth rate reduction due to FlucSM expression in HG medium (Figure 4A; Figure 4—figure supplement 1B). In contrast, negative controls using truncated Tom70_{cd} mutants that could not restore MP import did not produce the same growth defect (Figure 4—figure supplement 1C).

To further test whether the reduction in growth rate during proteotoxic stress was associated with impaired mitochondrial fitness, we assessed mitochondrial membrane potential (MMP) using the dye tetramethylrhodamine methyl ester (TMRM). In HG medium and after 90 min induction of FlucSM, there was a negative relationship between spGFP accumulation and MMP: spGFP-positive cells exhibited a significantly reduced MMP level than spGFP-negative cells (Figure 4C). Again, this difference was not observed in cells that grew in LG, whereas Tom70_{cd} overexpression led to a significant increase in the fraction of spGFP-positive cells with reduced MMP in both HG and LG medium (Figure 4B and C). These results suggest that Snf1 activation under glucose limitation protects mitochondrial and cellular fitness from FlucSM-associated proteotoxic stress.

Many neurodegenerative disease-associated aggregation-prone proteins, such as α -synuclein (13), FUS^{P525L} (14, 56), TDP-43 (15), amyloid beta (57), and C9ORF72-associated poly(GR) dipeptide (58), are detected in mitochondria of human patients or disease models and impair mitochondrial functions. We wonder whether such toxic effects of disease-associated proteins can be counteracted by AMPK activation. First, we used the spGFP reporter in yeast and observed mitochondrial import of α -synuclein and FUS^{P525L} in HG medium (Figure 4D and 4E; Figure 4—figure supplement 1F and G; Movie S2). We found that Snf1 activation via glucose limitation or $\Delta reg1$ significantly reduced their accumulation in mitochondria, whereas Tom70_{cd} overexpression reversed this effect (Figure 4D–I; Figure 4—figure supplement 1H–K). Mitochondrial import of α -synuclein and FUS^{P525L} in HG medium was associated with lower MMP, and Tom70_{cd} overexpression significantly increased the fraction of spGFP-positive and MMP-low cells in both HG and LG medium (Figure 4J–K). Furthermore, accumulation of α -

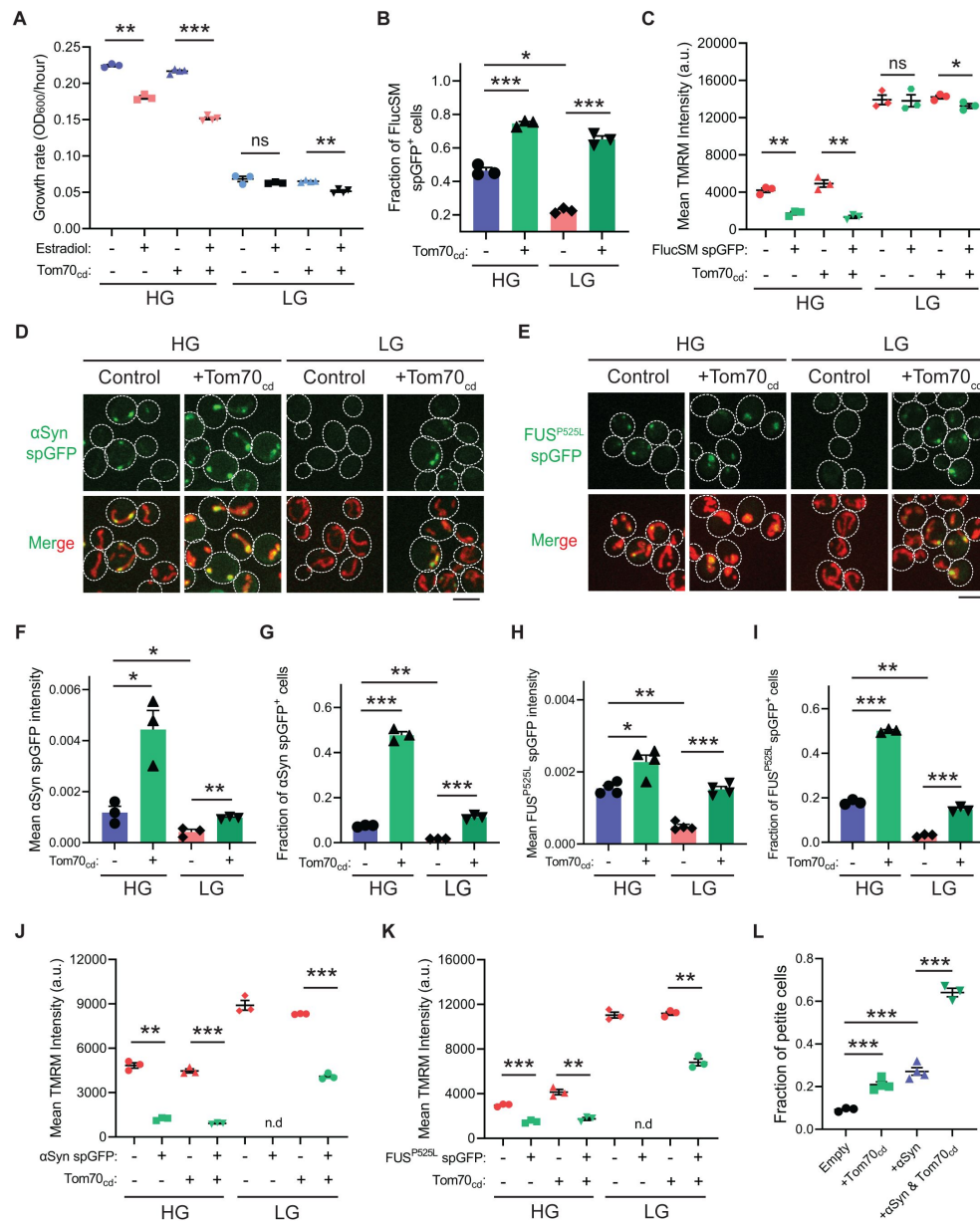


Figure 4.

Snf1 activation protects cellular fitness against proteotoxic stress.

(A) Growth rates of wild-type cells and cells overexpressing Tom70_{cd} with (Estradiol) or without (EtOH) FlucSM expression in HG and LG medium. Means ± SEM of OD₆₀₀ or growth rates are shown. Paired two-tailed *t*-test. (B) Fraction of FlucSM spGFP-positive cells measured by flow cytometry. Means ± SEM are shown. Unpaired two-tailed *t*-test for cells growing in the same medium. (C) Comparisons of mitochondrial membrane potential between FlucSM spGFP-negative and spGFP-positive cells measured by TMRM. Means ± SEM are shown. Paired two-tailed *t*-test. (D-I) Representative images and quantifications of α-synuclein (αSyn) spGFP and FUS^{P525L} spGFP signal. Shown in (F, H): means ± SEM of spGFP intensity measured by confocal imaging. Shown in (G, I): means ± SEM of fraction of spGFP-positive cells measured by flow cytometry. Unpaired two-tailed *t*-test for cells growing in the same medium. Paired two-tailed *t*-test for control cells between HG and LG medium. (J, K) Comparisons of membrane potential between αSyn or FUS^{P525L} spGFP-negative and spGFP-positive cells measured by TMRM. Means ± SEM are shown. Paired two-tailed *t*-test. n.d.: not determined due to limited positive cell counts in control cells growing in LG medium. (L) Fraction of respiratory-deficient petite cells measured by using tetrazolium overlay. Means ± SEM are shown. Unpaired two-tailed *t*-test. HG: 2% glucose; LG: 0.1% glucose plus 3% glycerol. **P* < 0.05; ***P* < 0.01; ****P* < 0.001; ns, not significant, *P* > 0.05. Scale bars, 5 μm.

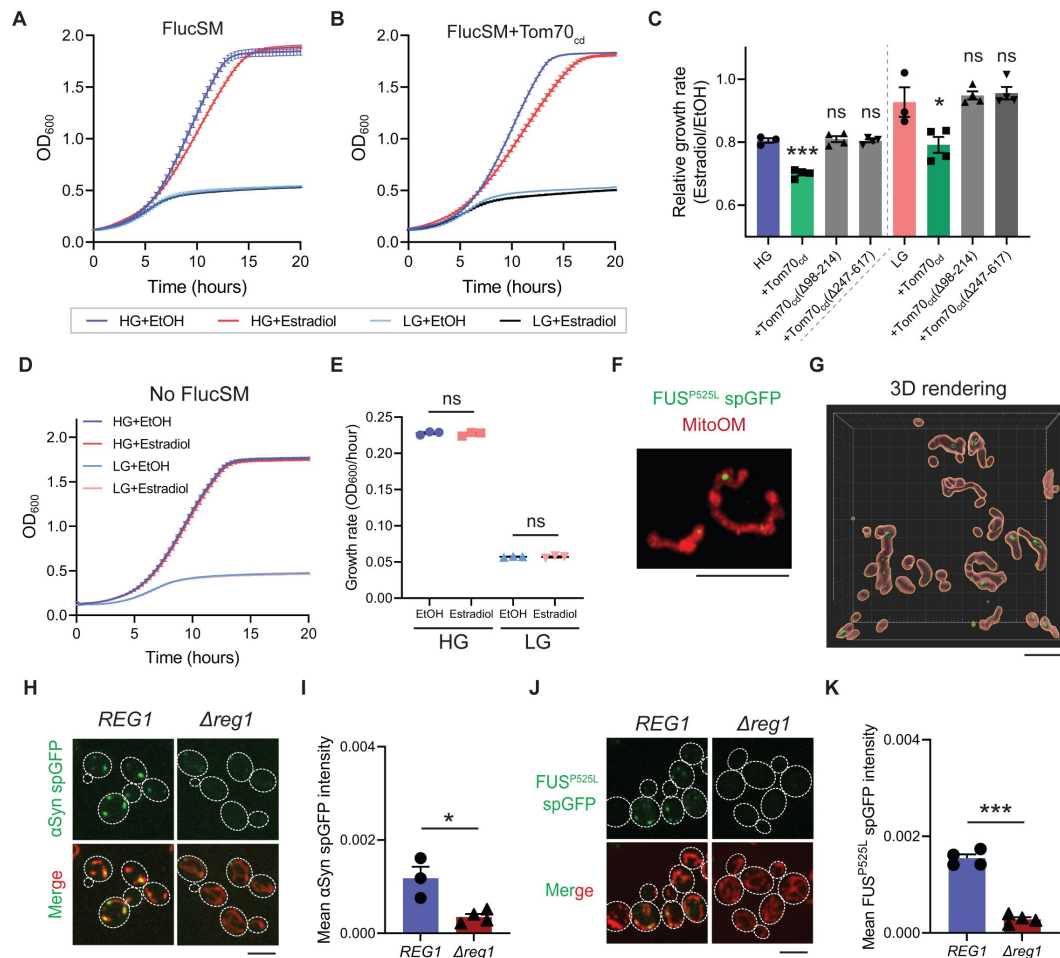


Figure 4—figure supplement 1.

Snf1 activation protects against stress associated with FlucSM overexpression and prevents the accumulation of α -synuclein and FUS^{P525L} in yeast mitochondria. (**A**, **B**) Growth curves of wild-type control cells (**A**) and cells overexpressing Tom70_{cd} (**B**) with (Estradiol) or without (EtOH) FlucSM expression in HG and LG medium. Fitted growth rates are shown in **Figure 4A**. (**C**) Relative growth rates of control cells and cells overexpressing Tom70_{cd} variants. Means $\pm$ SEM are shown. Unpaired two-tailed *t*-test between control and Tom70_{cd} variants that grew in the same medium (HG or LG). (**D**, **E**) Growth curves (**D**) and growth rates (**E**) of BY4741 cells lack of FlucSM spGFP reporter in HG and LG medium. Means $\pm$ SEM are shown. Paired two-tailed *t*-test. (**F**, **G**) Representative super-resolution imaging for FUS^{P525L} spGFP signal in mitochondria after 100 min of estradiol induction. Shown in (**F**): maximum projection image. Shown in (**G**): 3D rendered image. Mitochondrial OM: Mitochondrial outer membrane labeled with Tom70-mCherry. (**H**–**K**) Representative images (**H**, **J**) and quantification (**I**, **K**) of α -synuclein (α Syn) spGFP signal FUS^{P525L} spGFP signal in wild-type (*REG1*) and Δ *reg1* cells. Shown in (**I**, **K**): means $\pm$ SEM of spGFP intensity. Unpaired two-tailed *t*-test. HG: 2% glucose; LG: 0.1% glucose plus 3% glycerol. EtOH: ethanol. **P* < 0.05; ***P* < 0.01; ns, not significant, *P* > 0.05. Scale bars, 5 μ m.

synuclein in mitochondria correlated with a loss of respiratory capacity, as overexpression of Tom70_{cd} and α -synuclein synergistically promoted the formation of respiration-deficient petite cells (**Figure 4L** [↗](#)).

We next tested whether reducing mitochondrial accumulation of FUS^{P525L} ameliorates its cellular toxicity in human cells. FUS^{P525L} has been shown to bind mitochondrial Hsp60 and ATP synthase β -subunit to induce mitochondrial fragmentation and cell death ([14](#) [↗](#), [56](#) [↗](#)). We expressed FUS^{P525L} into human RPE-1 cells by transient transfection and confirmed the import of FUS^{P525L} into mitochondrial matrix using the spGFP reporter (**Figure 5A and B** [↗](#)). FUS^{P525} expression also caused the loss of MMP and elevated cell death compared to GST control (**Figure 5C and D** [↗](#)). Importantly, mitochondrial accumulation and fitness decline caused by FUS^{P525} expression were significantly reduced by activation of AMPK via AICAR treatment (**Figure 5B-D** [↗](#)). These results suggest a protective role of AMPK in FUS-induced cellular toxicities possibly through preventing the import of the disease protein into mitochondria.

Discussion

Metabolic imbalance and loss of proteostasis are interconnected hallmarks of aging and age-related diseases ([3](#) [↗](#), [5](#) [↗](#), [59](#) [↗](#)). Various metabolic signaling pathways, such as TOR, AMPK, Sirtuins, and insulin/IGF-1, sense metabolic stimuli, regulate cellular stress responses and influence major cytosolic protein quality control mechanisms including ubiquitin-proteasome pathway and autophagy ([59](#) [↗](#)). Mitochondria, the central target of metabolic signaling and major hub of energy production, participate in proteostasis by importing of cytosolic misfolded proteins lacking canonical mitochondrial targeting sequences via the MAGIC pathway ([10](#) [↗](#)). Here, our unbiased genetic screen in yeast revealed an unexpected link between cellular metabolism and proteostasis through MAGIC. Our data established Snf1/AMPK as a key regulator of misfolded protein import, which balances the mitochondrial metabolic and proteostasis functions in response to glucose availability and protects mitochondrial fitness under proteotoxic stress (**Figure 5E** [↗](#)). We speculate that, when glucose level is high and cells rely on glycolysis for ATP production, mitochondria play a ‘moonlighting role’ in cellular proteostasis through MAGIC, a process dependent on mitochondrial import and proteostasis machineries including chaperones, mitochondrial translocons, and proteases ([10](#) [↗](#)). On the other hand, when glucose is limited and cells rely on oxidative phosphorylation for ATP generation, Snf1/AMPK activation shuts down MAGIC and promotes import of essential mitochondrial preproteins, thus ensuring mitochondrial fitness and energy production.

The downstream mechanism of this regulation remains to be fully elucidated. We propose that in yeast Snf1 activates the Hap4-dependent expression of mitochondrial preproteins which could compete with MPs for limited TOM complexes under glucose-limiting condition. Using cytosolic domain of Tom receptors to dampen preprotein import, we showed that only Tom70_{cd} rescued MP import under Snf1 activation. A recent study ([60](#) [↗](#)) suggests that overexpression of full-length Tom70 leads to transcriptional activation for mitochondrial biogenesis. Whether the cytosolic Tom70_{cd} fragment plays an indirect role in mitochondrial import through transcriptional regulation should be tested in the future. Since Snf1/Hap4 activation elevates the expression of hundreds of mitochondrial preproteins ([35](#) [↗](#), [43](#) [↗](#)-[46](#) [↗](#)), it remains to be determined if specific preproteins or cytosolic factors are directly involved in inhibiting MP import. Furthermore, whether this metabolic control of MP import applies to other uncharacterized MAGIC substrates awaits further investigation.

Our data also suggest that the trimeric form of the TOM complex maintained by Tom6 is important for limiting MP entry under glucose restriction. We speculate that the receptor-binding state and substrate selectivity of different TOM conformations ([54](#) [↗](#)) could affect the permeability for MPs to enter mitochondria. Existing proteomic data suggest that the abundance of Tom6 is unaffected

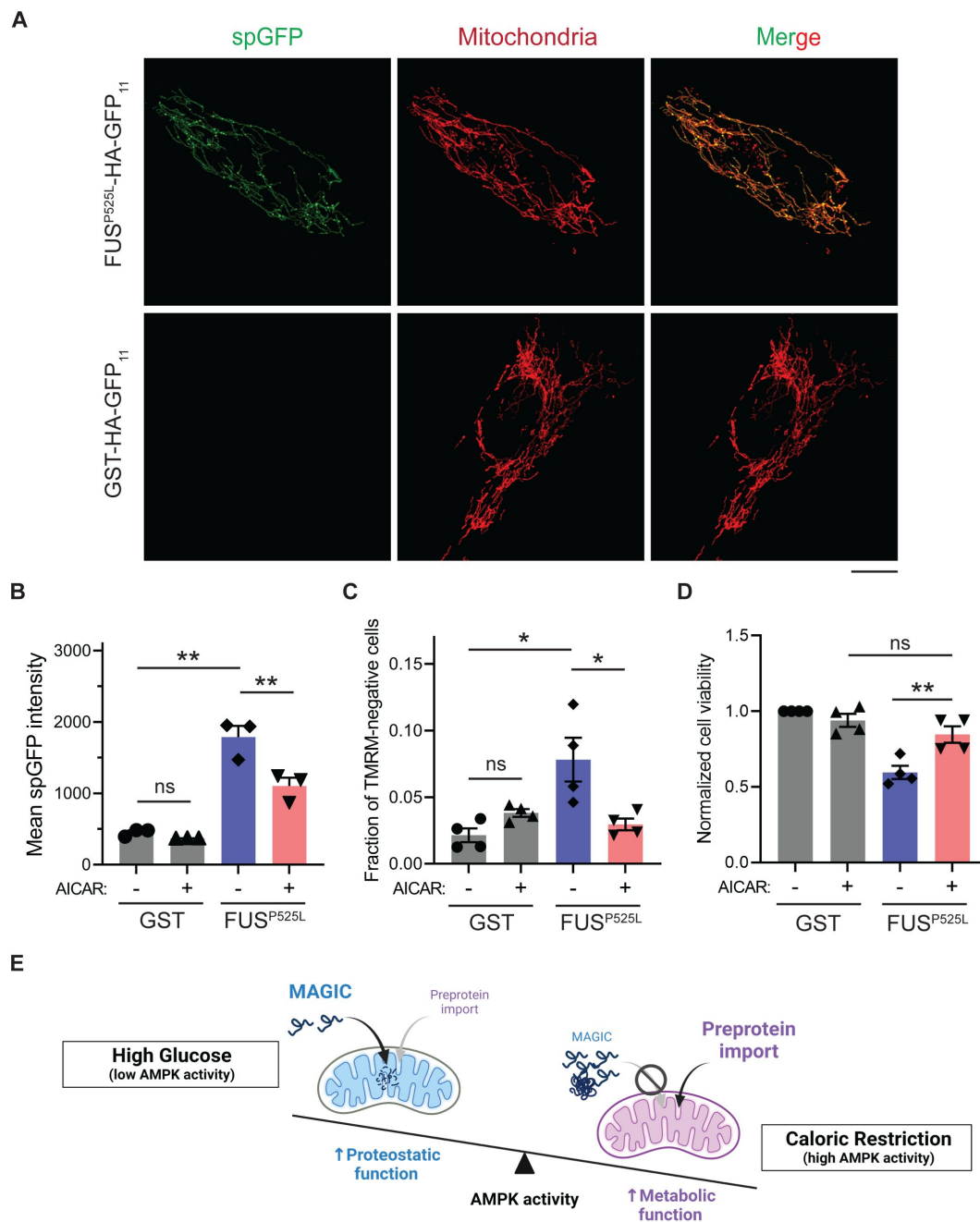


Figure 5.

AMPK activation prevents the accumulation of ALS-associated FUS^{P525L} in mitochondria of RPE-1 cells and alleviates FUS-induced cytotoxicity.

(A, B) Representative images (A) and flow cytometry quantification (B) of FUS^{P525L} spGFP and GST spGFP in mitochondria of RPE-1 cells treated with or without AICAR. Shown in (B): Means $\pm$ SEM of spGFP intensity. (C, D) Fraction of TMRM-negative cells (C) and normalized cell viability (D) of RPE-1 cells expressing GST-HA-GFP₁₁ or FUS^{P525L}-HA-GFP₁₁ with or without AICAR treatment. Means $\pm$ SEM are shown. (E) Working model wherein Snf1/AMPK balances the metabolic and proteostasis function of mitochondria in response to glucose availability. Paired two-tailed *t*-test for the same cell line treated with drug or control medium. Unpaired two-tailed *t*-test between cell lines expressing GST and FUS^{P525L}. **P* < 0.05; ***P* < 0.01; ns, not significant, *P* > 0.05. Scale bars, 10 μ m.

by Snf1 activation (35) (Figure 3A). As Tom6 can be phosphorylated by Cdk1 in a cell-cycle dependent manner (55), it may be interesting to investigate if Tom6 or other TOM complex components are targets of Snf1 kinase activity to directly modulate substrate specificity of the TOM complex.

A question raised by our findings is whether MAGIC is beneficial or detrimental to cells. Our data suggest that under physiological stress-free conditions, MP import and degradation in mitochondria is well-tolerated, but an acute or chronic increase in the cytosolic misfolded protein load could overwhelm mitochondrial proteostasis capacity leading to organellar damage. If so, the regulation of MAGIC by AMPK could help explain the beneficial effect of caloric restriction on life span extension in model organisms (43, 61). In humans, the role of AMPK in health and diseases is complex and not fully understood (62–64). While AMPK activity and mitochondrial gene expression mediated by downstream transcriptional factors such as PGC-1 α and FOXO are elevated during health-benefitting activities such as exercise (64), hyperactivated AMPK has also been reported in several neurodegenerative diseases with proteostasis decline (62). Our findings suggest that elevating AMPK activity may be beneficial for alleviating proteotoxicity associated with degenerative diseases. Further studies using genetic approaches and relevant *in vivo* models could help elucidate the physiological role of AMPK in balancing proteostasis and mitochondrial fitness.

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Author Contributions

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Methodology: YW, LR, JZ, AC-CC, XZ, AT

Investigation: YW, LR, RL

Visualization: YW, LR

Writing: YW, LR, JZ, RL

Supervision: RL

Funding acquisition: RL

Competing Interests

The authors declare no competing interests.

Data and materials availability

All data needed to evaluate the conclusions are available in the main text or the supplementary materials. Additional data and materials related to this study can be requested from the authors. Requests for resources and reagents should be directed to and will be fulfilled by the corresponding author, RL (rong@jhu.edu). The custom Python code for quantification of spGFP fluorescence can be found within the git repository at <https://github.com/RongLiLab/Wang-et-al.-2022.git> (10.7554/eLife.87518.2).

Supplementary Materials

Table S1 [↗](#): List of validated MAGIC regulators

Table S2 [↗](#): List of yeast strains and plasmids

Movie S1: 3D reconstructed SIM images showing FlucSM spGFP inside mitochondria after 90 min estradiol treatment. Mitochondrial outer membrane is labeled with Tom70-mCherry.

Movie S2: 3D reconstructed SIM images showing FUS^{P525L} spGFP inside mitochondria after 100 min estradiol treatment. Mitochondrial outer membrane is labeled with Tom70-mCherry.

Materials and Methods

Yeast strains, plasmids and culture media

Yeast strains used in this study are based on the BY4741 strain background and listed in **Table S2** [↗](#). Gene deletion and protein tagging were performed through PCR-mediated homologous recombination (65 [↗](#)) and verified by PCR genotyping. MAGIC YKO collection was constructed by incorporating MTS-mCherry-GFP₁₋₁₀ under GPD promoter into the TRP1 locus and tagging endogenous Lsg1 with GFP₁₁ in the YKO collection (18 [↗](#)). *Δreg1* and YKO strains harboring the deletion of the transcriptional factor downstream of Snf1 were freshly made and validated for at least three independent colonies.

Human α-synuclein tagged with GFP11 under GPD promoter was cloned and inserted into the *ura3Δ0* locus. FlucSM-HA-GFP₁₁ and FUS^{P525L}-HA-GFP₁₁ under GAL1 promoter were cloned from plasmids from our previous study (10 [↗](#)) and plasmid 416Gal-FUS-P525L-YFP, a gift from Aaron Gitler (Addgene plasmid #29628). FlucWT-HA-GFP₁₁ and FlucDM-HA-GFP₁₁ plasmids were constructed using site-directed mutagenesis kit (NEB) based on FlucSM-HA-GFP₁₁. Both GFP₁₁-

tagged Fluc proteins and GEM transcriptional factor (cloned from pJW1663, Addgene plasmid #112037) were stably integrated into yeast genome. GFP₁₋₁₀ was fused with the mitochondrial matrix protein Grx5 under GPD promoter, except in experiments involving *PIM1* or *pim1*^{S974D} mutant and α -synuclein spGFP where GFP₁₋₁₀ was fused to the C-terminus of endogenous Grx5 to avoid signal saturation. Wild-type *PIM1* or *pim1*^{S974D} mutant under CUP1 promoter, *HAP4*, cytosolic domain of Tom20 (1-97 aa), Tom22 (38-617aa), Tom70 (38-617aa) and truncated variants of Tom70cd under GPD promoter were cloned and stably integrated into yeast genome. Mitochondrial outer membrane was labeled with Tom70-mCherry or Tom70- RFP, except for the Tom70/71 deletion experiments in which mitochondria were labeled with mCherry-Fis1TM (8 [DOI](#)).

MAGIC YKO library construction, flow cytometry, and imaging during high-throughput screen were performed with synthetic defined minus histidine (SD-His) medium. Synthetic complete (SC) supplemented with 2% glucose (HG), 0.1% glucose plus 3% glycerol (LG), or 0.1% glucose (LG-Gly) was used for confocal imaging, luciferase assays, biochemistry, and TMRM staining. YEP medium (yeast extract-peptone) supplemented with 2% glucose (HG) or 0.1% glucose plus 3% glycerol (LG) was used for growth assays. Optical density at 600 nm (OD₆₀₀) was used to estimate the amount of yeast cells used in the various experiments.

Drug treatments

β -estradiol (E2758, MilliporeSigma) was dissolved in H₂O and added at a final concentration of 1 μ M. CuSO₄ (C1297, MilliporeSigma) was dissolved in H₂O and added at a final concentration of 0.5 mM. D-luciferin potassium salt (LUCK, GoldBio) was freshly dissolved in appropriate yeast media at a final concentration of 0.5 mM. Dorsomorphin (S7840, Selleck Chemicals; 11967, Cayman Chemical) dissolved in DMSO was added to RPE-1 cells at the final concentration of 10 μ M for 24 hours (66 [DOI](#)). AICAR was dissolved in DMSO (S1802, Selleck Chemicals) and added at the final concentration of 2 mM for 48 hours in the FlucDM experiment, or dissolved directly in media at the concentration of 2 mM (10010241, Cayman Chemical) for the FUS^{P525L} experiment (67 [DOI](#)). MG132 (C2211, MilliporeSigma) was dissolved in DMSO and added to YEP-based medium at a final concentration of 80 μ M.

Yeast library construction and genome-wide screen

MAGIC YKO was constructed with a two-step transformation using the Frozen-EZ Yeast Transformation II Kit (T2001, Zymo Research) following the microscale protocol in 96-well format. First, knockout strains were grown to saturation in deep-well plates containing 1 ml of YPD broth with G418 (200 μ g/mL, Corning). 150 μ L of refreshed mid-log phase cultures and 0.2 μ g of MTS-mCherry-GFP₁₋₁₀-clonNat DNA were used in the transformation setup on the epMotion 5075 liquid handling workstation (Eppendorf). To optimize transformation efficiency, the transformation mixtures were incubated for 2 hours and at the end of transformation they were transferred into deep-well plates with 4 volumes of YPD for 2 hours of outgrowth at 30 °C. The transformants were selected for 4-5 days in 1 ml of YPD broth with clonNAT (200 μ g/mL, GoldBio), resulting in the intermediate MTS-mCherry-GFP₁₋₁₀-clonNat library. Then the Lsg1-HA-GFP₁₁ tagging PCR product was integrated into the genome of the intermediate strains following the same protocol, with the exception that the final library was selected in SD-His medium.

Total 4645 YKO strains with Lsg1 spGFP reporter were cultured in 96-well plates, and spGFP intensities before and after heat shock (30 min at 42 °C) were measured at 488 nm excitation with appropriate filters on Attune NxT flow cytometer equipped with an auto sampler (Thermo). After subtracting background from the populational mean spGFP intensity, KOs displaying different spGFP pattern were determined by a cutoff (smaller than 1.1-fold increase after heat shock) and further validated by live cell confocal imaging. Based on the phenotype of mitochondrial spGFP intensity of each mutant at two imaging time points, Class 1 mutants were determined by the *P* value of comparing the spGFP/mCherry ratio of each single cell between KO and WT at permissive

temperature, $P < 0.01$, and Class 2 mutants were determined by the P value of comparing the spGFP intensity of each single cell of before and after heat shock for the same mutant, $P > 0.01$. Genes involved in known mitochondrial import pathways were excluded from analysis.

Confocal microscopy and imaging conditions

Live cell images were acquired using a Yokogawa CSU-10 spinning disc on the side port of a Carl Zeiss 200 m inverted microscope or a Carl Zeiss LSM-780 confocal system. Laser 488 or 561 nm excitation was applied to excite GFP or mCherry, respectively, and the emission was collected through the appropriate filters onto a Hamamatsu C9100-13 EMCCD on the spinning disc confocal system or the single-photon avalanche photodiodes on the Zeiss 780 system. Regarding the multi-track acquisition, the configuration of alternating excitation was used to avoid the bleed-through of GFP (for dual color imaging, GFP or mCherry labeled controls were applied for laser and exposure settings). The spinning disc and the LSM780 were equipped with a 100×1.45 NA Plan-Apochromat objective and a 63×1.4 oil Plan-Apochromat objective, respectively. For yeast 3D imaging, 0.5 μm step size for 6 μm in total in Z; for human cells, 1 μm step size. Images were acquired using MetaMorph (version 7.0, MDS Analytical Technologies) on the CSU-10 spinning disc system and Carl Zeiss ZEN software on the LSM780.

Yeast culture condition for imaging: yeast cells were cultured in SC or SD-His with appropriate carbon source overnight at 30 °C. The cells were then refreshed in the corresponding medium for at least 3 hours at 30 °C until reaching an OD₆₀₀ of about 0.2. For estradiol-GEM inducible systems, 1 μM of β -estradiol was added to the medium for 90 min unless indicated otherwise. For copper-inducible overexpression of *PIM1* or its mutant, 0.5 mM CuSO₄ was added for 2 hours, followed by the estradiol induction for 2 hours. All images in the same experiments were acquired with the same laser and exposure settings. Image processing was performed using ImageJ software (NIH) or Imaris software (Oxford Instruments Group). For visualization purposes, images were scaled with bilinear interpolation and shown as the maximum projection on Z for fluorescent channels. Cell boundaries were delineated according to white-field images.

Split-GFP quantification

Split-GFP fluorescence from confocal images was quantified by using a custom Python code described previously ([10](#)). In brief, mCherry and GFP intensities were summed along the z-axis, and then subjected to a random walk segmentation of the background and watershed segmentation of adjoining cells. For each cell, the mCherry channel was thresholded at 5% of maximal value to detect mitochondria, and median GFP intensity within mitochondria was calculated as spGFP intensity per cell. In the YKO imaging validation, Lsg1 spGFP/mCherry ratio of each cell was used for statistical analyses. For Lsg1 spGFP signal detected in Δsnf1 , Δltv1 , and *WT* cells, populational means spGFP/mCherry of at least 3 biological repeats were calculated. Adjusting Lsg1 spGFP intensity to mitochondrial mCherry intensity avoided the potential effect of changing local abundance of GFP₁₋₁₀ on Lsg1 spGFP signal after heat shock. For estradiol-inducible systems that did not involve heat shock, populational mean spGFP intensity of each biological repeat was used for the following analyses. For the flow cytometry quantification, populational mean GFP intensities of at least 25,000 single cells were calculated for the following analyses. Most quantifications were shown as absolute intensity values with an arbitrary unit. Normalized spGFP intensities were calculated to highlight the relative changes between different strains.

Mammalian cell culture, transfection, imaging and quantification

Human RPE-1 cells were cultured in Dulbecco's Modified Eagle Medium: Nutrient Mixture F-12 (DMEM/F12) (GIBCO), supplemented with 10% (v/v) fetal bovine serum (FBS), 100 IU/ml penicillin. Transient transfections were performed with Lipofectamine 3000 (Invitrogen) according to the manufacturer's instructions. RPE-1 cells were dually transfected with MTS-mCherry-GFP₁₋₁₀ and the protein of interest tagged with GFP₁₁ (2.5 μg of each plasmid was applied). For imaging,

MatTek (P35G-0-14-C) dish was used to culture cells, and cells were located using the mCherry channel only. Cells were imaged or analyzed by flow cytometry after 24 or 48 hours of transfection for FUS^{P525L} or FlucDM, respectively. For flow cytometry analysis of FUS^{P525L} spGFP system, cells were permeabilized with digitonin buffer (0.32 M sucrose, 5 mM CaCl₂, 3 mM Mg[Acetate]₂, 0.1 mM EDTA, 10 mM Tris-HCl, 100 µg/ml digitonin) for 8- 10 min, in order to remove spGFP signal outside of mitochondria in cytosol.

To evaluate cell death caused by FUS^{P525L} overexpression, equal number of RPE-1 cells were seeded in 6-well plates and transfected with GST or FUS^{P525L}, with or without AICAR. Compared to GST transfection control, FUS^{P525L} resulted in significant floating dead cells. Number of attached cells after 24 hours of transfection were analyzed with Attune NxT flow cytometer as a proxy for cell viability.

Cell lysates, immunoblots and antibodies

For yeast experiments, 1-2 ml of yeast cells in the indicated background and medium was collected by centrifugation and snap frozen in liquid nitrogen for storage. Pellets were disrupted, boiled in 120 µl 1X LDS sample buffer (Thermo) for 10 min, and vortexed with an equal volume of 0.5 mm acid-washed glass beads to break cells at 4 °C for 2 min with a 1 min interval. Cell lysates were boiled for 5 min, separated from glass beads by 15,000 g centrifugation at room temperature for 30 sec, and analyzed by SDS-PAGE. For mammalian data, RPE-1 cells were washed with PBS and lysed with RIPA buffer (MilliporeSigma) supplemented with protease inhibitors on ice for 20-30 min. Cell lysates were further sonicated and incubated on ice for 5 min, followed by 10 min 21,200 g centrifugation at 4 °C. The supernatant was collected and analyzed by SDS-PAGE.

Transfer was performed using iBlot2 (Thermo) and immunoblots were developed using Clarity Western ECL substrate (Bio-Rad) for HRP-linked secondary antibodies, or directly using fluorescent IRDye secondary antibodies (LI-COR). Images were acquired by using LI-COR imaging systems and analyzed in Image Studio (LI-COR). HA-tag (C29F4) rabbit mAb #3724 from Cell Signaling Technology. PGK1 mouse mAb (22C5D8) from Invitrogen. FLAG mouse clone M2 (F1804) from MilliporeSigma. GFP Living Colors A.v. mAb clone JL-8 (632381) from Takara Bio.

Firefly luciferase assays

Firefly luciferase assays in yeast were carried out as described previously ([36](#)). In brief, after 90 min of estradiol induction, 100 µl of cells was vigorously mixed with 100 µl of 1 mM D-luciferin in a white 96-well plate (655073, Greiner Bio-One), and light emission was immediately measured by the luminescence detection mode in Cytation 5 (Biotek). Luciferase activities were normalized to cell density measured by OD₆₀₀ and adjusted to total abundance of FlucSM protein measured by immunoblotting.

Mig1 nucleocytoplasmic translocation

The nucleocytoplasmic distribution of Mig1-GFP was quantified using a custom ImageJ macro and MATLAB script as described previously ([68](#)). In brief, nuclear protein Pus1-RFP was used to create nucleoplasmic mask for each cell ([69](#)). Cytoplasm was defined by a dilated nuclear mask ([68](#)). The nuclear-cytoplasmic ratio of each cell was calculated by dividing the mean nuclear intensity by the mean cytoplasmic intensity. Populational mean nuclear-cytoplasmic ratio of at least 3 biological replicates were used for statistical analyses.

Yeast growth curve

Yeast cells with indicated genetic background were cultured in corresponding media. Overnight cultures were refreshed for 4 hours at 30 °C and the OD₆₀₀ of the cells was measured and adjusted to 0.05. Diluted cell suspension was added to a 96-well plate with 2 µM estradiol or ethanol as control. The wells along the perimeter of the plate were pre-filled with 200 µL cell-free medium to

prevent evaporation. The OD₆₀₀ was continuously monitored at 30 °C using Citation 5 (Biotek) every 20 min with constant shaking. Data were extracted and analyzed using the R package GroFit (<https://cran.r-project.org/src/contrib/Archive/grofit/>) (70).

Mitochondrial membrane potential measurements

Yeast cells expressing MPs and growing in appropriate medium was collected, incubated with 2.5 μM TMRM (21437, Cayman Chemical) for 15 min at 30 °C and washed twice by fresh medium before recording with Attune NxT flow cytometer equipped with appropriate filter sets. A SpGFP intensity threshold was applied so that less than 1% of cells displayed positive spGFP in the ethanol-treated control groups with no expression of MPs. Mean TMRM intensities of at least 25,000 cells were calculated for each biological replicate.

RPE-1 cells transfected with either GST or FUS^{P525L} for 24 hours were washed once with PBS and added with complete media containing 150 nM TMRM for 30 min at 37 °C. After incubation, cells were washed with PBS and trypsinized into single cells. Cell suspensions were pelleted and re-suspended in PBS for analysis on the Attune NxT flow cytometer.

Tetrazolium overlay assay

Yeast tetrazolium overlay was performed to measure the respiratory deficiency in a yeast population as previously described (71). In brief, yeast cells were inoculated in YPD media at 30 °C overnight. Around 100 cells were plate on YPD plates and grew for 4 days at 30 °C. The tetrazolium test medium consists of 1.5% agar and 0.1% tetrazolium (17342, Cayman Chemical) in 0.067 M phosphate buffer at pH 7.0. Test was performed by pouring 15 ml of melted test medium at 55 °C over a YPD plate. The number of large red colonies (respiration-sufficient) and small white colonies (respiration-deficient) were counted after 1 hour of incubation at 30 °C.

Super resolution imaging

Structured illumination microscopy (SIM) images were acquired with a GE OMX-SR Super-Resolution Microscope 3D Structure Illumination (3D-SIM) equipped with high-sensitivity PCO sCMOS cameras, or LSM880-Airyscan FAST Super-Resolution microscopy equipped with 63×/1.4 PlanApo oil. GFP and mCherry were excited with 488 nm and 568 nm lasers, respectively. The SIM images were reconstructed with the Softworx and aligned following the Applied Precision protocols, and Zeiss images were reconstructed with Airyscan processing. 3D rendering was performed with Imaris (Oxford Instruments Group).

Statistical analysis

Descriptions of statistical tests and *P* values can be found in Figure Legends. At least three biological replicates were analyzed in all experiments. Statistical analyses were performed with GraphPad Prism 6.0 and Microsoft Excel. No statistical methods were used to predetermine the sample size. The experiments were not randomized, and the investigators were not blinded to allocation during experiments and outcome assessment.

Systematic Name	Standard Name	MAGIC Phenotype
YDR477W	SNF1	Class 1
YML016C	PPZ1	Class 1
YJR120W		Class 1
YOL055C	THI20	Class 1
YKL057C	NUP120	Class 1
YML024W	RPS17A	Class 2
YDR083W	RRP8	Class 2
YCR002C	CDC10	Class 2
YKL143W	LTV1	Class 2
YLL026W	HSP104	Class 2
YPR159W	KRE6	Class 2
YOR096W	RPS7A	Class 2
YMR116C	ASC1	Class 2
YPR057W	BRR1	Class 2
YJR074W	MOG1	Class 2
YCR068W	ATG15	Class 2
YML062C	MFT1	Class 2
YML026C	RPS18B	Class 2
YML013W	UBX2	Class 2
YMR032W	HOF1	Class 2
YNR029C	ZNG1	Class 2
YDL020C	RPN4	Class 2
YER151C	UBP3	Class 2
YMR255W	GFD1	Class 2
YMR307W	GAS1	Class 2
YOR035C	SHE4	Class 2
YOL072W	THP1	Class 2
YDL083C	RPS16B	Class 2
YOR258W	YOR258W	Class 2
YOL129W	VPS68	Class 2
YHR163W	SOL3	Class 2
YLR372W	ELO3	Class 2
YKL191W	DPH2	Class 2
YIR032C	DAL3	Class 2
YBR020W	GAL1	Class 2
YJR145C	RPS4A	Class 2
YDR085C	AFR1	Class 2
YGR019W	UGA1	Class 2
YEL068C		Class 2

Table S1

List of validated MAGIC regulators

(Highlighted in yellow: ribosome-associated genes based on KEGG)

YIL112W	HOS4	Class 2
YKL198C	PTK1	Class 2
YER087C-A		Class 2
YJL200C	ACO2	Class 2
YJL160C	PIR5	Class 2
YMR034C	RCH1	Class 2
YGR132C	PHB1	Class 2
YLL033W	IRC19	Class 2
YGR072W	UPF3	Class 2
YGR016W		Class 2
YCR071C	IMG2	Class 2
YER060W	FCY21	Class 2
YER075C	PTP3	Class 2
YGR129W	SYF2	Class 2
YPR146C		Class 2
YEL012W	UBC8	Class 2
YJR113C	RSM7	Class 2
YPL173W	MRPL40	Class 2
YDL057W		Class 2
YBR068C	BAP2	Class 2
YHR200W	RPN10	Class 2
YOR298C-A	MBF1	Class 2
YER056C	FCY2	Class 2
YNL081C	SWS2	Class 2
YGL114W	YGL114W	Class 2
YAR030C		Class 2
YLR053C	NRS1	Class 2
YMR089C	YTA12	Class 2
YBR058C	UBP14	Class 2
YBR175W	SWD3	Class 2
YBR231C	SWC5	Class 2
YDR073W	SNF11	Class 2
YDR115W	MRX14	Class 2
YGR136W	LSB1	Class 2
YGR159C	NSR1	Class 2
YHL033C	RPL8A	Class 2
YHR011W	DIA4	Class 2
YHR143W	DSE2	Class 2
YCL005W	LDB16	Class 2
YCL037C	SRO9	Class 2
YLR131C	ACE2	Class 2
YMR074C	SDD2	Class 2
YKL009W	MRT4	Class 2
YKL128C	PMU1	Class 2

Table S1 (continued)

YKL132C	RMA1	Class 2
YGR056W	RSC1	Class 2
YOR125C	CAT5	Class 2
YAL043C-a		Class 2
YLL015W	BPT1	Class 2
YOR235W	IRC13	Class 2
YJL179W	PFD1	Class 2
YLR387C	REH1	Class 2
YLR388W	RPS29A	Class 2
YDR173C	ARG82	Class 2
YGL197W	MDS3	Class 2
YGL194C	HOS2	Class 2
YGL210W	YPT32	Class 2
YPL049C	DIG1	Class 2
YGL085W	LCL3	Class 2
YNL156C	NSG2	Class 2
YKL213C	DOA1	Class 2
YKR042W	UTH1	Class 2
YKR057W	RPS21A	Class 2
YLR065C	SND2	Class 2
YIL043C	CBR1	Class 2
YIL049W	DFG10	Class 2
YIL088C	AVT7	Class 2
YIL054W		Class 2
YOL111C	MDY2	Class 2
YOL122C	SMF1	Class 2
YER091C	MET6	Class 2
YNL316C	PHA2	Class 2
YDL213C	NOP6	Class 2
YDR006C	SOK1	Class 2
YDR025W	RPS11A	Class 2
YBR297W	MAL33	Class 2
YCR025C		Class 2
YML088W	UFO1	Class 2
YNL008C	ASI3	Class 2
YNL010W	PYP1	Class 2
YNR047W	FPK1	Class 2
YBR027C		Class 2
YBR043C	QDR3	Class 2
YML036W	CGI121	Class 2
YPL004C	LSP1	Class 2
YML066C	SMA2	Class 2
YBR133C	HSL7	Class 2
YDL002C	NHP10	Class 2

Table S1 (continued)

YBR172C	SMY2	Class 2
YDL021W	GPM2	Class 2
YDR462W	MRPL28	Class 2
YDR500C	RPL37B	Class 2
YGL136C	MRM2	Class 2
YER174C	GRX4	Class 2
YER167W	BCK2	Class 2
YMR221C	FMP42	Class 2
YIL094C	LYS12	Class 2
YGR254W	ENO1	Class 2
YMR257C	PET111	Class 2
YMR278W	PRM15	Class 2
YMR291W	TDA1	Class 2
YMR303C	ADH2	Class 2
YNL303W		Class 2
YNL302C	RPS19B	Class 2
YNL265C	IST1	Class 2
YNL264C	PDR17	Class 2

Table S1 (continued)

Strain ID	Genotype	Source	
BY4741	<i>MATa his3Δ1; leu2Δ0; met15Δ0; ura3Δ0</i>		
RLY8616	<i>GRX5-GFP₁₁-His3MX6; trp1::P_{GPD}-MTS-mCherry-GFP₁₋₁₀-natMX6</i>	Ruan, L. et al., 2017.	
RLY8618	<i>LSG1-GFP₁₁-His3MX6; trp1::P_{GPD}-MTS-mCherry-GFP₁₋₁₀-natMX6</i>	Ruan, L. et al., 2017.	
RLY9798	<i>LSG1-GFP₁₁-His3MX6; trp1::P_{GPD}-MTS-mCherry-GFP₁₋₁₀-natMX6; Δsnf1::kanMX6</i>	This study	
RLY9799	<i>LSG1-GFP₁₁-His3MX6; trp1::P_{GPD}-MTS-mCherry-GFP₁₋₁₀-natMX6; Δltv1::kanMX6</i>	This study	
RLY9800	<i>ura3Δ0::GEM-hphMX6; trp1::P_{GPD}-GRX5-HA-GFP₁₋₁₀-natMX6; HO::P_{GAL1}-FlucSM-HA-GFP₁₁-His3MX6; TOM70-mCherry-Ura3MX6</i>	This study	
RLY9801	<i>ura3Δ0::GEM-hphMX6; trp1::P_{GPD}-GRX5-HA-GFP₁₋₁₀-natMX6; HO::P_{GAL1}-FlucSM-HA-GFP₁₁-His3MX6; TOM70-mCherry-Ura3MX6; Δreg1::Leu2</i>	This study	
RLY9802	<i>trp1::P_{GPD}-GRX5-HA-GFP₁₋₁₀-natMX6; amp::GEM-P_{GAL1}-FlucSM-HA-GFP₁₁-kanMX6; TOM70-RFP-hphMX6</i>	This study	
RLY9803	<i>trp1::P_{GPD}-GRX5-HA-GFP₁₋₁₀-natMX6; amp::GEM-P_{GAL1}-FlucSM-HA-GFP₁₁-kanMX6; TOM70-mCherry-Ura3MX6</i>	This study	
RLY9804	<i>trp1::P_{GPD}-GRX5-HA-GFP₁₋₁₀-natMX6; amp::GEM-P_{GAL1}-FlucWT-HA-GFP₁₁-kanMX6; TOM70-mCherry-Ura3MX6</i>	This study	
RLY9805	<i>trp1::P_{GPD}-GRX5-HA-GFP₁₋₁₀-natMX6; amp::GEM-P_{GAL1}-FlucDM-HA-GFP₁₁-kanMX6; TOM70-mCherry-Ura3MX6</i>	This study	
RLY9806	<i>MIG1-GFP-His3MX6; PUS1-RFP-hphMX6</i>	This study	
RLY9807	<i>MIG1-GFP-His3MX6; PUS1-RFP-hphMX6; Δreg1::Leu2</i>	This study	
RLY9808	<i>ura3Δ0::P_{CUP1}-PIM1-Ura3; GRX5-GFP₁₋₁₀-natMX6; trp1::P_{GPD}-mCherry-Fis1TM-hphMX6; amp::GEM-P_{GAL1}-FlucSM-HA-GFP₁₁-kanMX6</i>	This study	
RLY9809	<i>ura3Δ0::P_{CUP1}-pim1^{S974D}-Ura3; GRX5-GFP₁₋₁₀-natMX6; trp1::P_{GPD}-mCherry-Fis1TM-hphMX6; amp::GEM-P_{GAL1}-FlucSM-HA-GFP₁₁-kanMX6</i>	This study	
RLY9810	<i>trp1::P_{GPD}-MTS-mCherry-natMX6; amp::GEM-P_{GAL1}-FlucSM-HA-GFP₁₁-kanMX6</i>	This study	
RLY9811	<i>trp1::P_{GPD}-MTS-mCherry-natMX6; amp::GEM-P_{GAL1}-FlucSM-HA-GFP₁₁-kanMX6; Δreg1::His3MX6</i>	This study	
RLY9812	<i>ura3Δ0::GEM-hphMX6; trp1::P_{GPD}-GRX5-HA-GFP₁₋₁₀-natMX6; HO::P_{GAL1}-FlucSM-HA-GFP₁₁-His3MX6; TOM70-mCherry-Ura3MX6; Δreg1::Leu2; Δhap4::kanMX6</i>	This study	
RLY9813	<i>trp1::P_{GPD}-GRX5-HA-GFP₁₋₁₀-natMX6; amp::GEM-P_{GAL1}-FlucSM-HA-GFP₁₁-kanMX6</i>	This study	

Table S2

List of yeast strains and plasmids

	<i>TOM70-mCherry-Ura3MX6; HO::P_{GPD}-HAP4-hphMX6</i>		
RLY9814	<i>ura3Δ0::GEM-hphMX6; trp1::P_{GPD}-GRX5-HA-GFP₁₋₁₀-natMX6; HO::P_{GAL1}-FlucSM-HA-GFP₁₁-His3MX6; TOM70-mCherry-Ura3MX6; Δatg1::kanMX6</i>	This study	
RLY9815	<i>ura3Δ0::GEM-hphMX6; trp1::P_{GPD}-GRX5-HA-GFP₁₋₁₀-natMX6; HO::P_{GAL1}-FlucSM-HA-GFP₁₁-His3MX6; TOM70-mCherry-Ura3MX6; Δatg15::kanMX6</i>	This study	
RLY9816	<i>ura3Δ0::GEM-hphMX6; trp1::P_{GPD}-GRX5-HA-GFP₁₋₁₀-natMX6; HO::P_{GAL1}-FlucSM-HA-GFP₁₁-His3MX6; TOM70-mCherry-Ura3MX6; Δmig1::kanMX6</i>	This study	
RLY9817	<i>ura3Δ0::GEM-hphMX6; trp1::P_{GPD}-GRX5-HA-GFP₁₋₁₀-natMX6; HO::P_{GAL1}-FlucSM-HA-GFP₁₁-His3MX6; TOM70-mCherry-Ura3MX6; Δmig2::Leu2</i>	This study	
RLY9818	<i>ura3Δ0::GEM-hphMX6; trp1::P_{GPD}-GRX5-HA-GFP₁₋₁₀-natMX6; HO::P_{GAL1}-FlucSM-HA-GFP₁₁-His3MX6; TOM70-mCherry-Ura3MX6; Δmig1::kanMX6; Δmig2::Leu2</i>	This study	
RLY9819	<i>ura3Δ0::GEM-hphMX6; trp1::P_{GPD}-GRX5-HA-GFP₁₋₁₀-natMX6; HO::P_{GAL1}-FlucSM-HA-GFP₁₁-His3MX6; TOM70-mCherry-Ura3MX6; Δreg1::Leu2; Δcat8::kanMX6</i>	This study	
RLY9820	<i>ura3Δ0::GEM-hphMX6; trp1::P_{GPD}-GRX5-HA-GFP₁₋₁₀-natMX6; HO::P_{GAL1}-FlucSM-HA-GFP₁₁-His3MX6; TOM70-mCherry-Ura3MX6; Δreg1::Leu2; Δsip4::kanMX6</i>	This study	
RLY9821	<i>ura3Δ0::GEM-hphMX6; trp1::P_{GPD}-GRX5-HA-GFP₁₋₁₀-natMX6; HO::P_{GAL1}-FlucSM-HA-GFP₁₁-His3MX6; TOM70-mCherry-Ura3MX6; Δreg1::Leu2; Δrds2::kanMX6</i>	This study	
RLY9822	<i>ura3Δ0::GEM-hphMX6; trp1::P_{GPD}-GRX5-HA-GFP₁₋₁₀-natMX6; HO::P_{GAL1}-FlucSM-HA-GFP₁₁-His3MX6; TOM70-mCherry-Ura3MX6; Δreg1::Leu2; Δadr1::kanMX6</i>	This study	
RLY9823	<i>trp1::P_{GPD}-GRX5-HA-GFP₁₋₁₀-natMX6; amp::GEM-P_{GAL1}-FlucSM-HA-GFP₁₁-kanMX6; TOM70-mCherry-Ura3MX6; HO::P_{GPD}-tom20_{cd}-hphMX6</i>	This study	
RLY9824	<i>trp1::P_{GPD}-GRX5-HA-GFP₁₋₁₀-natMX6; amp::GEM-P_{GAL1}-FlucSM-HA-GFP₁₁-kanMX6; TOM70-mCherry-Ura3MX6; HO::P_{GPD}-tom22_{cd}-hphMX6</i>	This study	
RLY9825	<i>ura3Δ0::GEM-hphMX6; trp1::P_{GPD}-GRX5-HA-GFP₁₋₁₀-natMX6; HO::P_{GAL1}-FlucSM-HA-GFP₁₁-His3MX6; TOM70-mCherry-Ura3MX6; amp::P_{GPD}-tom70_{cd}-3xFLAG-kanMX6</i>	This study	

Table S2 (continued)

RLY9826	<i>ura3Δ0::GEM-hphMX6; trp1::P_{GPD}-GRX5-HA-GFP₁₋₁₀-natMX6; HO::P_{GAL1}-FlucSM-HA-GFP₁₁-His3MX6; TOM70-mCherry-Ura3MX6; amp::P_{GPD}-tom70_{cd}(Δ98-214)-3xFLAG-kanMX6</i>	This study	
RLY9827	<i>ura3Δ0::GEM-hphMX6; trp1::P_{GPD}-GRX5-HA-GFP₁₋₁₀-natMX6; HO::P_{GAL1}-FlucSM-HA-GFP₁₁-His3MX6; TOM70-mCherry-Ura3MX6; amp::P_{GPD}-tom70_{cd}(Δ247-617)-3xFLAG-kanMX6</i>	This study	
RLY9828	<i>ura3Δ0::GEM-hphMX6; trp1::P_{GPD}-GRX5-HA-GFP₁₋₁₀-natMX6; HO::P_{GAL1}-FlucSM-HA-GFP₁₁-His3MX6; trp1::P_{GPD}-mCherry-Fis1TM-kanMX6</i>	This study	
RLY9829	<i>ura3Δ0::GEM-hphMX6; trp1::P_{GPD}-GRX5-HA-GFP₁₋₁₀-natMX6; HO::P_{GAL1}-FlucSM-HA-GFP₁₁-His3MX6; trp1::P_{GPD}-mCherry-Fis1TM-kanMX6; Δtom70::Ura3MX6</i>	This study	
RLY9830	<i>ura3Δ0::GEM-hphMX6; trp1::P_{GPD}-GRX5-HA-GFP₁₋₁₀-natMX6; HO::P_{GAL1}-FlucSM-HA-GFP₁₁-His3MX6; trp1::P_{GPD}-mCherry-Fis1TM-kanMX6; Δtom71::Leu2</i>	This study	
RLY9831	<i>ura3Δ0::GEM-hphMX6; trp1::P_{GPD}-GRX5-HA-GFP₁₋₁₀-natMX6; HO::P_{GAL1}-FlucSM-HA-GFP₁₁-His3MX6; trp1::P_{GPD}-mCherry-Fis1TM-kanMX6; Δtom70::Ura3MX6; Δtom71::Leu2</i>	This study	
RLY9832	<i>ura3Δ0::GEM-hphMX6; trp1::P_{GPD}-GRX5-HA-GFP₁₋₁₀-natMX6; HO::P_{GAL1}-FlucSM-HA-GFP₁₁-His3MX6; TOM70-mCherry-Ura3MX6; amp::P_{GPD}-tom70_{cd}-3xFLAG-kanMX6; Δtom6::Leu2</i>	This study	
RLY9833	<i>ura3Δ0::GEM-hphMX6; trp1::P_{GPD}-GRX5-HA-GFP₁₋₁₀-natMX6; HO::P_{GAL1}-FlucSM-HA-GFP₁₁-His3MX6</i>	This study	
RLY9834	<i>ura3Δ0::P_{GPD}-a-Synuclein-HA-GFP₁₁-His3MX6; GRX5-GFP₁₋₁₀-natMX6; TOM70-mCherry-Ura3MX6; amp::P_{GPD}-tom70_{cd}-3xFLAG-kanMX6</i>	This study	
RLY9835	<i>ura3Δ0::P_{GPD}-a-Synuclein-HA-GFP₁₁-His3MX6; GRX5-GFP₁₋₁₀-natMX6; TOM70-mCherry-Ura3MX6</i>	This study	
RLY9836	<i>ura3Δ0::P_{GPD}-a-Synuclein-HA-GFP₁₁-His3MX6; GRX5-GFP₁₋₁₀-natMX6; TOM70-mCherry-Ura3MX6; Δreg1::Leu2</i>	This study	
RLY9837	<i>ura3Δ0::P_{GPD}-a-Synuclein-HA-GFP₁₁-His3MX6; GRX5-GFP₁₋₁₀-natMX6; TOM70-mCherry-Ura3MX6; trp1::P_{GPD}-tom70_{cd}-3xFLAG-kanMX6</i>	This study	
RLY9838	<i>ura3Δ0::P_{GPD}-a-Synuclein-HA-GFP₁₁-His3MX6; GRX5-GFP₁₋₁₀-natMX6</i>	This study	
RLY9839	<i>ura3Δ0::P_{GPD}-a-Synuclein-HA-GFP₁₁-His3MX6; GRX5-GFP₁₋₁₀-natMX6; trp1::P_{GPD}-tom70_{cd}-3xFLAG-kanMX6</i>	This study	

Table S2 (continued)

RLY9840	<i>trp1::P_{GPD}-GRX5-HA-GFP₁₋₁₀-natMX6; TOM70-mCherry-Ura3MX6; amp::GEM-P_{GAL1}-FUS^{P525L}-HA-GFP₁₁-kanMX6</i>	This study	
RLY9841	<i>ura3Δ0::GEM-hphMX6; trp1::P_{GPD}-GRX5-HA-GFP₁₋₁₀-natMX6; HO::P_{GAL1}-FUS^{P525L}-HA-GFP₁₁-His3MX6; TOM70-mCherry-Ura3MX6</i>	This study	
RLY9842	<i>ura3Δ0::GEM-hphMX6; trp1::P_{GPD}-GRX5-HA-GFP₁₋₁₀-natMX6; HO::P_{GAL1}-FUS^{P525L}-HA-GFP₁₁-His3MX6; TOM70-mCherry-Ura3MX6; Δreg1::Leu2</i>	This study	
RLY9843	<i>ura3Δ0::GEM-hphMX6; trp1::P_{GPD}-GRX5-HA-GFP₁₋₁₀-natMX6; HO::P_{GAL1}-FUS^{P525L}-HA-GFP₁₁-His3MX6; TOM70-mCherry-Ura3MX6; amp::P_{GPD}-tom70_{cd}-3xFLAG-kanMX6</i>	This study	
RLY9844	<i>trp1::P_{GPD}-GRX5-HA-GFP₁₋₁₀-natMX6; amp::GEM-P_{GAL1}-FUS^{P525L}-HA-GFP₁₁-kanMX6</i>	This study	
RLY9845	<i>trp1::P_{GPD}-GRX5-HA-GFP₁₋₁₀-natMX6; amp::GEM-P_{GAL1}-FUS^{P525L}-HA-GFP₁₁-kanMX6; trp1::P_{GPD}-tom70_{cd}-3xFLAG-kanMX6</i>	This study	
RLY9846	<i>ura3Δ0::GEM-hphMX6; trp1::P_{GPD}-GRX5-HA-GFP₁₋₁₀-natMX6; HO::P_{GAL1}-FlucSM-HA-GFP₁₁-His3MX6; TOM70-mCherry-Ura3MX6; Δpdr5::kanMX6</i>	This study	
RLY9847	<i>ura3Δ0::GEM-hphMX6; trp1::P_{GPD}-GRX5-HA-GFP₁₋₁₀-natMX6; HO::P_{GAL1}-FlucSM-HA-GFP₁₁-His3MX6; TOM70-mCherry-Ura3MX6; Δhap4::kanMX6</i>	This study	
Plasmid ID	Construct	Vector type	Source
RLB918	<i>TRP1::P_{GPD}-MTS-mCherry-GFP₁₋₁₀-natMX6</i>	Yeast expression	Ruan, L. et al., 2017.
RLB919	<i>TRP1::P_{GPD}-Grx5-HA-GFP₁₋₁₀-natMX6</i>	Yeast expression	Ruan, L. et al., 2017.
pJW1663	<i>GEM-P_{GAL1}-GFP-kanMX6</i>	Yeast expression	Costa, E.A. et al. 2018.
RLB1050	<i>TRP1::P_{GPD}-mCherry-Fis1TM-KanMX6</i>	Yeast expression	Ruan, L. et al., 2017.
RLB1051	<i>GEM-P_{GAL1}-FlucSM-HA-GFP₁₁-KanMX6</i>	Yeast expression	This study
RLB1052	<i>GEM-P_{GAL1}-FlucWT-HA-GFP₁₁-KanMX6</i>	Yeast expression	This study
RLB1053	<i>GEM-P_{GAL1}-FlucDM-HA-GFP₁₁-KanMX6</i>	Yeast expression	This study
RLB1054	<i>GEM-P_{GAL1}-FUS^{P525L}-HA-GFP₁₁-KanMX6</i>	Yeast expression	This study
RLB1055	<i>pRS316-P_{CUP1}-PIM1-Ura3</i>	Yeast expression	Nitika et al. 2022
RLB1056	<i>pRS316-P_{CUP1}-pim1^{S974D}-Ura3</i>	Yeast expression	Nitika et al. 2022

Table S2 (continued)

RLB1057	<i>pRS313-HO(homology)-P_{GAL1}-FlucSM-HA-GFP₁₁-His3MX6-HO(homology)</i>	Yeast expression	This study
RLB1058	<i>pRS313-HO(homology)-P_{GAL1}-FUS^{P525L}-HA-GFP₁₁-His3MX6-HO(homology)</i>	Yeast expression	This study
RLB1059	<i>pRS316-HO(homology)-P_{GPD}-HAP4-hphMX6-HO(homology)</i>	Yeast expression	This study
RLB1060	<i>TRP1::P_{GPD}-tom70_{cd}-3xFLAG-KanMX6</i>	Yeast expression	This study
RLB1061	<i>TRP1::P_{GPD}-tom70_{cd}(Δ98-214)-3xFLAG-KanMX6</i>	Yeast expression	This study
RLB1062	<i>TRP1::P_{GPD}-tom70_{cd}(Δ247-617)-3xFLAG-KanMX6</i>	Yeast expression	This study
RLB1063	<i>pRS316-HO(homology)-P_{GPD}-tom20_{cd}-hphMX6-HO(homology)</i>	Yeast expression	This study
RLB1064	<i>pRS316-HO(homology)-P_{GPD}-tom22_{cd}-hphMX6-HO(homology)</i>	Yeast expression	This study
RLB1065	<i>P_{GPD}-α-Synuclein-HA-GFP₁₁-His3MX6</i>	Yeast expression	This study
RLB1066	<i>P_{CMV}-FUS^{P525L}-HA-GFP₁₁</i>	Mammalian expression	Ruan, L. et al., 2017.
RLB912	<i>P_{CMV}-MTS-mCherry-GFP₁₋₁₀</i>	Mammalian expression	Ruan, L. et al., 2017.
RLB914	<i>P_{CMV}-FlucDM-HA-GFP₁₁</i>	Mammalian expression	Ruan, L. et al., 2017.
RLB916	<i>P_{CMV}-GST-HA-GFP₁₁</i>	Mammalian expression	Ruan, L. et al., 2017.

Table S2 (continued)

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Editors

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Joint Public Review:

In this study, Wang et al extend on their previous finding of a novel quality control pathway, the MAGIC pathway. This pathway allows misfolded cytosolic proteins to become imported into mitochondria and there they are degraded by the LON protease. Using a screen, they identify Snf1 as a player that regulates MAGIC. Snf1 inhibits mitochondrial protein import via the transcription factor Hap4 via an unknown pathway. This allows cells to adapt to metabolic changes, upon high glucose levels, misfolded proteins become imported and degraded, while during low glucose growth conditions, import of these proteins is prevented, and instead import of mitochondrial proteins is preferred.

This is a nice and well-structured manuscript reporting on important findings about a regulatory mechanism of a quality control pathway. The findings are obtained by a combination of mostly fluorescent protein-based assays. Findings from these assays support the claims well.

While this study convincingly describes the mechanisms of a mitochondria-associated import pathway using mainly model substrates, my major concern is that the physiological relevance of this pathway remains unclear: what are endogenous substrates of the pathway, to which extent are they imported and degraded, i.e. how much does MAGIC contribute to overall misfolded protein removal (none of the experiments reports quantitative "flux" information). Lastly, it remains unclear by which mechanism Snf1 impacts on MAGIC or whether it is "only" about being outcompeted by mitochondrial precursors.

Author Response

The following is the authors' response to the original reviews.

Reviewer #1 (Public Review):

"MAGIC" was introduced by the Rong Li lab in a Nature letters article in 2017. This manuscript is an extension of this original work and uses a genome wide screen the Baker's yeast to decipher which cellular pathways influence MAGIC. Overall, this manuscript is a logical extension of the 2017 study, however the manuscript is challenging to follow, complicated by the data often being discussed out of sequence. Although the manuscripts make claims of a mechanism being pinpointed, there are many gaps and the true mechanisms of how the factors identified in the screen influence MAGIC is not clear. A key issue is that there are many assumptions drawn on previous literature, but central aspects of the mechanisms being proposed are not adequately shown.

Key comments:

- 1. Reasoning and pipelines presented in the first two sections of the results are disordered and do not follow figure order. In some instances, the background to experimental analyses such as detailing the generation of spGFP constructs in the YKO mutant library, or validation of Snf1 activation are mentioned after respective results are discussed. This needs to be fixed.*

We thank the reviewer for pointing out potential confusion to readers. We have revised the first two sections according to reviewer's suggestion. (Page 4-6)

- 1. In general there is a lack of data to support microscopy data and supporting quantification analysis. The validity of this data could be significantly strengthened with accompanying western blots showing accumulation of a given constructs in mitochondrial sub compartments (as was the case in the lab's original paper in 2017).*

We appreciate the reviewer's suggestion on biochemical validations. However, the validity of this imaging-based assay for detecting import of cytosolic misfolded proteins into mitochondria, including the use of FlucSM as a model misfolding-prone protein, was carefully established in our previous study by using appropriate controls, super resolution imaging, APEX-based proximity labeling, and classical biochemical fractionation and protease protection assay (Ruan et al., 2017 Nature, ref. 10). We have reminded readers of these validation experiments in the previous study on Page 4, line 14-17.

In recent years, advancements in imaging-based tools have allowed many protein interactions and dynamic processes, which were previously examined by using biochemical assays in lysates of populations of cells, to be observed with various level of quantitation in live cells with intact cellular compartments. Many of these assays, e.g., the RUSH assay for ER to Golgi transport, FRAP-based analysis for nuclear/cytoplasmic shuttling of proteins, or FRET-based assays for protein-protein interactions, have been well accepted and even embraced by the respective fields of study once validated with genetic and biochemical approaches. The advantages for live-cell imaging-based assays are often their unique ability to report dynamic processes or unstable molecular species with spatiotemporal sensitivity. Respectfully, it is our view, based on our own experience, that the traditional protease protection assay is not adequate or sufficiently quantitative for examining the presence of unstable misfolded proteins in mitochondrial sub-compartments, given the obligatorily lengthy in vitro cell lysis and mitochondrial isolation process, during which the unstable proteins are continuously being degraded. This likely explains our previous biochemical fractionation result that only weak protein signals were detected in the matrix fraction (Ruan et al., 2017 Nature, ref. 10). In addition, unlike stably folded, native mitochondrial matrix proteins, misfolded/unfolded proteins such as Lsg1 or FlucSM are highly susceptible to protease treatment. This sensitivity makes the assay unreliable for detecting such proteins if trace amount of the protease penetrates mitochondrial membranes during cell lysis even without detergent treatment.

While we agree that protease protection assay is highly valuable for qualitative detection of the presence of a protein in certain mitochondrial compartments or determining its topology on membranes, this assay (regrettably in our hands) does not allow quantitative comparisons that were necessary for this study, because of inherent sample to sample variation, yet the laborious and low throughput nature of this assay makes it difficult for adequate statistical analysis. Furthermore, the level of protein detection in various fractions is highly sensitive to how the sample is treated with protease and detergent. Our imaging-based quantification, on the other hand, allows us to compare increased or decreased presence of GFP11-tagged proteins in mitochondria under different metabolic conditions or in different mutant or wild-type strains. Data from hundreds of cells and at least three independent biological replicates allowed us to apply adequate statistical analysis to aid our conclusion.

1. Much of the mechanisms proposed relies on the Snf1 activation. This is however not shown but assumed to be taking place. Given that this activation is central to the mechanism proposed, this should be explicitly shown here - for example survey the phosphorylation status of the protein.

Both REG1 deletion and low glucose conditions have been demonstrated extensively for Snf1 phosphorylation and activation in yeast (e.g., many seminal papers from Marian Carlson's and other lab, such as ref. 24-28). In our study, we have indeed corroborated this by showing that Mig1 was exported from the nucleus in Δ reg1 mutant and in low glucose conditions (Figure 1—figure supplement 2H and I. The mechanism of Snf1-mediated nuclear export of Mig1 has been characterized in detail as well (e.g., ref. 29-31).

Recommendations for the authors: please note that you control which, if any, revisions, to undertake

Reviewer #1 (Recommendations For The Authors):

SPECIFIC COMMENTS

Genetic Screen o Line 20 - the narrative moves to SNF1, but the reasoning for the selection of this Class I substrate is not defined. What was the basis for this selection -

what happened to the other Class I substrates. It is stated in the text that the other Class I proteins show the same increase in spGFP signal. The data showing this should be included in the Supp Figure 1 for transparency.

We have moved the narratives of Snf1 function to the second section and clarified that we were interested in this gene due to its central role in metabolism and mitochondrial functions that may influence MAGIC (Page 5: line 16-20). Other genes in class 1 were shown in Table S1. Detailed discussion of other genes in this category is beyond the scope of this study.

Snf1/AMPK prevents MP accumulation in mitochondria:

The FlucDM data in human RPE-1 mitochondria seems to be added to only increase the significance of the work. The mechanisms suggested here with Hap4 would not be possible in human cells as there is no homologue of this protein in human cells. Making generalisations that these pathways are conserved based on this one experiment is not appropriate.

We appreciate this feedback. Although the focus of this study is the regulation of MAGIC by the yeast AMPK Snf1, we would like to share our initial observation that suggests a similar role of AMPK in human RPE-1 cells. We acknowledge that the underlying mechanisms regarding the downstream transcription factors and pathway for misfolded protein import could be different in mammalian cells, but the overall effect of AMPK in mitochondrial biogenesis is well known to resemble that of Snf1. To avoid making over-generalization, we changed our statement of conclusion to: 'These results suggest that AMPK in human cells regulates MP accumulation in mitochondria following a similar trend as in yeast, although the underlying mechanisms might differ between these organisms.' (Page 7: line 2-4)

Mechanisms of MAGIC regulation by Snf1:

While the lysosome is ruled out here the authors have not considered the proteasomes. Is there a reason for this? Given accumulation of aggregates outside of mitochondria, and previous connections of the proteasome to mitochondrial quality control this would be an obvious thing to check. We examined the role of lysosomal degradation here because it is known to be activated under Snf1 active condition (ref. 37). We appreciate this feedback and have included a new analysis on MG132 treated FlucSM spGFP strains in which PDR5 gene was deleted to avoid drug efflux.

This result suggests that the proteasome inhibitor did not ablate the difference in FlucSM accumulation between these conditions. That MG132 promoted mitochondrial accumulation of FlucSM in both high glucose and low glucose conditions was not surprising, as FlucSM is also degraded by proteasome in the cytosol (Ruan et al., 2017 Nature, ref. 10), and preventing this pathway could divert more of such protein molecules toward MAGIC. (Page 7: line 26-29).

Line 13 "we hypothesized that elevated expression of mitochondrial preproteins induced by the activation of Snf1-Hap4 axis (REF) may outcompete MPs for import channels". This statement has some assumptions. The authors have not shown that Snf1 is activated in their models and more importantly that they have an accumulation of mitochondrial preproteins. The data that follows using the cytosolic domains of the receptors is hard to rationalise without seeing evidence that there is in fact pre-protein accumulation or impacts on the mitochondrial proteome in this system.

As stated in our response to main point [3], Snf1 activation in reg1 mutant or in low glucose is evidenced by our data showing Mig1 export from nucleus to cytoplasm and had also been shown in many previous publications. A recent study (Tsuboi et al., 2020 eLife) also showed a dramatic increase in mitochondrial volume fraction in Δ reg1 cells and wild-type cells in

respiratory conditions, further supporting the role of Snf1 in mitochondrial biogenesis. We have provided relevant references in the manuscript (ref. 24-28).

The ability of Tom70 cytosolic domain (Tom70cd), which can bind mitochondrial preproteins but not localize to mitochondria due to lack of N-terminal targeting sequence, to compete with endogenous Tom70 for mitochondrial preproteins has been well documented (ref. 47-49). However, we agree with the reviewer that a future quantitative proteomics study to measure changes in mitochondrial proteome under Tom70cd over-expression could allow more accurate interpretation of our experimental result.

AMPK protects cellular fitness during proteotoxic stress:

The inhibition of preprotein import by overexpressing the cytosolic domains of receptors is not supported with some proof of principle data. If this was working as the authors assume, it is not clear why only an effect with Tom70 is observed. The majority of the mitochondrial proteome is imported via Tom20/Tom22 so this does not align with what the authors are suggesting. Is the Tom70CD and any associated Hsp proteins facilitating the observed changes to the MPs?

We thank the reviewer for raising this point. We expressed different TOM receptor cytosolic domains but found that Tom70cd had the strongest rescue on MAGIC under AMPK activation conditions. It is possible that certain Tom70 substrates or Tom70-associated heat shock proteins inhibit the import of MAGIC substrates. We admit that a clear explanation of this unexpected observation necessitates a better understanding of how native and MAGIC substrates are selected and imported by the outer-membrane channel. We can only offer our best interpretation based on the current state of the understanding, and we feel that we have been careful to acknowledge such in the manuscript.

While the effect of AMPK inactivation reducing FUS accumulation was striking, this was all in the context of overexpression and may not be physiologically relevant - or may occur very transiently under basal conditions. Is GST an appropriate control here, why not use WT FUS? Likewise, one representative image is shown in Figure 5 - can the authors show western blotting that mitochondrial accumulation of FUS can be reduced with AMPK activation?

We thank the reviewer for this suggestion, however, overexpressed FUS WT is also aggregation prone (Zhihui Sun et al., 2011, PloS Biology; Shulin Ju, 2011, PloS Biology; Jacqueline C. Mitchell et., 2013, Acta Neuro). We believe that GST, as a well-folded protein, is an appropriate control (Ruan et al., 2017 Nature, ref. 10). As we discussed in response to main point [1], the in vitro assay involving protease protection and western blots do not allow reliable quantitative comparison in our hands.

In text changes.

The analysis pipeline of the YKO mutant library should be introduced at the very start of the first paragraph, not the end.

Addressed on Page 4, second paragraph

"Fluc" should be introduced as "Firefly luciferase" within the first paragraph of the first section, also need to define SM and DM in FlucSM/FlucDM - these appear to be missing.

Addressed in both Introduction (Page 2: line 29; Page 3: line 8-9) and re-clarified in Result (Page 5: line 27-29)

The role of Reg1 should be explicitly stated in the text, not just in the figure.

Addressed on Page 6: line 3-6

Figure 1H legend states Reg1 (WT) is Snf1-inactive and Reg1 KO is Snf1-active. This wording is confusing and is not supported by data, but by assumption. If the authors want to use this wording then evidence needs to be provided - as suggested above.

We have changed this and other legends to only show genotypes and medium conditions.

"Tom70cd overexpression also exacerbated growth rate reduction due to FlucSM expression in HG medium (Figure 4A; Figure 4 - figure supplement 1A)" should be figure supplement 1B.

Fixed on Page 10: line 10

"These results suggest that glucose limitation protects mitochondria and cellular fitness during FlucSM induced proteotoxic stress through Snf1-dependent inhibition of MP import into mitochondria". The phrase "Snf1-dependent inhibition of MP import into mitochondria" may be misleading, as Snf1 isn't modulating import directly but is acting on transcriptional regulators to modulate mitochondrial import under stress.

We restated the conclusion as follows: 'These results suggest that Snf1 activation under glucose limitation protects mitochondrial and cellular fitness under FlucSM-associated proteotoxic stress.' (Page 10: line 20-

1.

"... Significantly increased the fraction of spGFP-positive and MMP-low cells in both HG and LG medium (Figure 4G-K)" should be (Figure 4J-K).

Fixed on Page 11: line 3

Reviewer #2 (Public Review):

Work of Rong Li's lab, published in Nature 2017 (Ruan et al, 2017), led the authors to suggest that the mitochondrial protein import machinery removes misfolded/aggregated proteins from the cytosol and transports them to the mitochondrial matrix, where they are degraded by Pim1, the yeast Lon protease. The process was named mitochondria as guardian in cytosol (MAGIC).

The mechanism by which MAGIC selects proteins lacking mitochondrial targeting information, and the mechanism which allows misfolded proteins to cross the mitochondrial membranes remained, however, enigmatic. Up to my knowledge, additional support of MAGIC has not been published. Due to that, MAGIC is briefly mentioned in relevant reviews (it is a very interesting possibility!), however, the process is mentioned as a "proposal" (Andreasson et al, 2019) or is referred to require "further investigation to define its relevance for cellular protein homeostasis (proteostasis)" (Pfanner et al, 2019).

Rong Li's lab now presents a follow-up story. As in the original Nature paper, the major findings are based on in vivo localization studies in yeast. The authors employ an aggregation prone, artificial luciferase construct (FlucSM), in a classical split-GFP assay: GFP1-10 is targeted to the matrix of mitochondria by fusion with the mitochondrial

protein Grx5, while GFP11 is fused to FlucSM, lacking mitochondrial targeting information. In addition the authors perform a genetic screen, based on a similar assay, however, using the cytosolic misfolding-prone protein Lsg1 as a read-out.

My major concern about the manuscript is that it does not provide additional information which helps to understand how specifically aggregated cytosolic proteins, lacking a mitochondrial targeting signal could be imported into mitochondria. As it stands, I am not convinced that the observed FlucSM-/Lsg1-GFP signals presented in this study originate from FlucSM-/Lsg1-GFP localized inside of the mitochondrial matrix. The conclusions drawn by the authors in the current manuscript, however, rely on this single approach.

In the 2017 paper the authors state: "... we speculate that protein aggregates engaged with mitochondria via interaction with import receptors such as Tom70, leading to import of aggregate proteins followed by degradation by mitochondrial proteases such as Pim1." Based on the new data shown in this manuscript the authors now conclude "that MP (misfolded protein) import does not use Tom70/Tom71 as obligatory receptors." The new data presented do not provide a conclusive alternative. More experiments are required to draw a conclusion.

In my view: to confirm that MAGIC does indeed result in import of aggregated cytosolic proteins into the mitochondrial matrix, a second, independent approach is needed. My suggestion is to isolate mitochondria from a strain expressing FlucSM-GFP and perform protease protection assays, which are well established to demonstrate matrix localization of mitochondrial proteins. In case the authors are not equipped to do these experiments I feel that a collaboration with one of the excellent mitochondrial labs in the US might help the MAGIC pathway to become established.

We thank Reviewer 2 for these suggestions, but we would like to respectfully offer our difference in opinion:

a. Regarding the suggestion “to isolate mitochondria from a strain expressing FlucSM-GFP and perform protease protection assays”, in our previous study (Ruan et al., 2017 Nature, ref. 10), we have indeed applied two independent biochemical approaches: APEX-mitochondrial matrix proximity labeling and classic protease protection assay using non-spGFP strains, both consistently confirmed the entry of misfolded proteins into mitochondria under proteotoxic stress. Our super-resolution imaging further confirmed the import of the split GFP-labeled proteins to be inside mitochondria. Moreover, as we discussed in response to Reviewer 1’s main point [2], while the suggested biochemical assay is useful for validating topology within mitochondria, it is not quantitative and may not reliably report the in vivo accumulation of misfolded proteins in mitochondria due to the isolation process that takes hours, during which the unstable proteins could be continuously degraded within mitochondria.

While we agree with the reviewer that we do not yet understand how misfolded proteins are imported into mitochondria, it would be unfair to state “as it stands, I am not convinced..” simply because the underlying mechanism remains to be elucidated. We would like to point out that targeting sequences for many well-established mitochondrial proteins are still not well defined. It is well known that mitochondrial targeting sequences are not as uniformly predictable as, for example, nuclear targeting sequences. Our finding that deletion of TOM6 enhances the import of misfolded proteins suggest that their import may involve the TOM channel in a more promiscuous conformation, which may reduce the requirement for a specific sequence-based targeting signal associated with the substrate.

b. Regarding the role of Tom70, in our 2017 study, using proteomics and subsequently immunoprecipitation we validated the binding, albeit not necessarily direct, between misfolded protein FlucSM and Tom70. Therefore, “we speculate that protein aggregates

engaged with mitochondria via interaction with import receptors such as Tom70". Recent studies from different labs confirmed the interactions between Tom70 and aggregation prone proteins (Backes et al., 2021, Cell Reports; Liu et al., 2023, PNAS). In the current study, surprisingly, knockout of TOM70 did not block MAGIC, suggesting redundant components of mitochondria import system may facilitate the recruitment of misfolded proteins in the absence of Tom70, and this does not contradict the notion that Tom70 helps tether protein aggregates to mitochondria.

c. Regarding other studies also showing the import of misfolding or aggregation-prone cytosolic proteins into mitochondria, there have been at least several recent studies in the literature for mammalian cells involving either model substrates or disease proteins (e.g., ref. 12-15; 56-58; Vicario, M. et al. 2019 Cell Death Dis.). The studies are briefly mentioned in Introduction (Page 3, paragraph 2). The present manuscript documents a major effort from our group using whole genome screen in yeast to understand the mechanism and regulation of MAGIC. Many of the screen hits have yet to be studied in detail. We full agree that much remains to be understood about whether and how this pathway affects proteostasis and what might be the evolutionary origin for such a mechanism.

Additional comments:

The genetic screen:

The genetic screen identified five class 1 deletion strains, which lead to enhanced accumulation of Lsg1GFP and a larger set of class 2 mutants, which lead to reduced accumulation. Please note, in my opinion it is not clear that accumulation of the reporters occurs inside the mitochondria. In any case, the authors selected one single protein for further analysis: Snf1, the catalytic subunit of the yeast SNF complex, which is required for respiratory growth of yeast.

The results of the screen are not discussed in any detail. The authors mention that ribosome biogenesis factors are abundant among class 2 mutants. Noteworthy, Lsg1 is involved in 60S ribosomal subunit biogenesis. As Lsg1-GFP11 is overexpressed in the screen this should be discussed. Class 2 mutants also include several 40S ribosomal subunit proteins (only one of the 60S subunit). What does this imply for the MAGIC model? Also, it should be discussed that the screen did not identify Δ reg1 and Δ hap4, which I had expected as hits based on the data shown in later parts of the manuscript.

We apologize for the confusion, but the GFP11 tag was in fact knocked into the C-terminus of Lsg1 in the endogenous LSG1 locus, and so Lsg1 was not overexpressed in the screen. We have made sure that this information is clearly conveyed in the revised manuscript (Page 4: line 20-22). How the ribosome small subunit affects MAGIC is beyond the focus of the current study and will be pursued in the future.

Regarding why certain mutants did not come out of our initial screen, this is not unexpected as the YKO collection, although extremely valuable to the community, is known to be potentially affected by false knockouts, suppressor accumulation and cross contamination (for references, e.g., Puddu et al., 2019 Nature). Additionally, high-through screens can also miss real hits. In our experience using this collection in several studies, we often found additional hits from analysis of genes implicated by known genetic or biochemical interactions.

Mutant yeast strains and growth assays:

The Δ reg1 strain grows poorly in all growth conditions and frequently accumulates extragenic suppressor mutations (Barrett et al, 2012). It would be good to make sure that this is not the case in the strains employed in this study. My suggestion is to do (and

show) standard yeast plating assays with the relevant mutant strains including $\Delta reg1$, $snf1$, $hap4$, $\Delta reg1\Delta hap4$ without the split GFP constructs and also with them (i.e. the strains that were used in the assays).

We thank the reviewer for the suggestion. We were indeed aware of potential accumulation of suppressor mutations from the YKO library. Therefore, deletion mutants like $\Delta reg1$ and loss of TFs downstream of Snf1 that we used in the study after the initial screen were all freshly made and validated. At least 3 independent colonies were analyzed for each mutant (mentioned in Methods & Materials; Page 33, line 57). Moreover, the plating assay suggested here may not reveal additional information other than growth, which was taken into consideration during our experiments.

Activation of Snf1 in the relevant strains should be tested with the commercially available antibody recognizing active Snf1, which is phosphorylated at Snf1-T210.

Snf1 activation was validated by the Mig1 exporting from the nucleus. We also noted above that many studies have clearly demonstrated Snf1 activation in $reg1$ mutant and under low glucose growth (e.g., ref. 24-28).

Effects of Snf1, Reg1, Hap4 and respiratory growth conditions:

The authors show that split GFP reporters show enhanced accumulation during fermentative growth, in $\Delta snf1$, and $\Delta reg1\Delta hap4$ and fail to accumulate during respiratory growth, in $\Delta reg1$ and upon overexpression of HAP4. Analysis of $\Delta hap4$ should be included in Fig. 2. The suggestion that upon activation of Snf1 enhanced Hap4-dependent expression "outcompetes" misfolded protein import seems unlikely as only a fraction of mitochondrial genes is under control of Hap4. Without further experimental evidence I do not find that a valid assumption. More likely, the membrane potential plays a role: it is low during fermentative growth, in $\Delta snf1$ and $\Delta reg1\Delta hap4$, and high during respiratory growth and in $\Delta reg1$ (Hübscher et al, 2016). Such an effect of the membrane potential seems to contradict the findings in the 2017 paper and the issue should be clarified and discussed. In any case, these data do not reveal that GFP reporters accumulate inside of the mitochondria. Based on the currently available evidence they may accumulate in close proximity/attached to the mitochondria. This has to be tested directly (see above).

We have included our analysis of $\Delta hap4$ in Page 8: line 14-15 and Figure 2—figure supplement 1H. Consistent with our result for $\Delta reg1\Delta hap4$ in glucose-rich medium, HAP4 deletion also resulted in a significant increase in mitochondrial accumulation of FlucSM in low glucose medium compared to WT. It did not have effect in high glucose condition in which Snf1 is largely inactive.

It is our view that the importance of Hap4 should not be judged by the number of nuclear encoded mitochondrial proteins they regulate. Still, this sub-group comprises a considerable number of proteins (at least 55 genes upregulated by Hap4 overexpression, ref. 43), and certain substrates may be more competitive with misfolded cytosolic proteins for import. Our genetic data strongly suggest that the inhibitory effect of active Snf1 on MAGIC is through Hap4, although we agree with the reviewer that detailed mechanism on how Hap4 substrates may compete with misfolded proteins need to be addressed in future studies.

Membrane potential is important for mitochondrial import. During respiratory growth and in $\Delta reg1$, membrane potential is well known to be elevated comparing to fermentative condition (e.g., Figure 4C). Our observation that the import of misfolded proteins into mitochondria is reduced under these conditions simply suggests that this reduction is not due to a lack of

membrane potential. This is not in any way contradictory to our 2017 finding that misfolded protein import requires membrane potential (ref. 10).

Again, the accumulation of misfolded proteins in mitochondria, especially the model protein FlucSM, has been validated by using super resolution imaging (Figure 1—figure supplement 1A) in addition to the protease protection assay in our 2017 study.

Introduction and Discussion:

Both are really short, too short in my view. Please provide some background of the general principals of mitochondrial protein import and information of how exactly translocation of cytosolic, aggregated proteins (lacking targeting information) is supposed to work. I do not understand exactly how the authors actually envisage the process.

We thank the reviewer for the suggestion. In the revised manuscript, we have extended both Introduction (Page 2-3) and Discussion section (Page 11-13)

The results from the 2022 eLife paper (Liu et al, 2022), which suggests that Tom70 may "regulate both the transcription/biogenesis and import of mitochondrial proteins so the nascent mitochondrial proteins do not compromise cytosolic proteostasis or cause cytosolic protein aggregation" should be discussed with regard to the data obtained with overexpression of the Tom70 soluble domain.

We thank the reviewer for pointing out that study and we have included a brief comment in Discussion section (Page 12: line 13-16). As the function of Tom70 appears to be complex, we cannot exclude the possibility that overexpression of the cytosolic domain has additional or indirect effects in addition to that due to preprotein binding.

Andreasson, C., Ott, M., and Buttner, S. (2019). Mitochondria orchestrate proteostatic and metabolic stress responses. EMBO Rep 20, e47865.

Barrett, L., Orlova, M., Maziarz, M., and Kuchin, S. (2012). Protein kinase A contributes to the negative control of Snf1 protein kinase in Saccharomyces cerevisiae. Eukaryot Cell 11, 119-128.

Hubscher, V., Mudholkar, K., Chiabudini, M., Fitzke, E., Wolfle, T., Pfeifer, D., Drepper, F., Warscheid, B., and Rospert, S. (2016). The Hsp70 homolog Ssb and the 14-3-3 protein Bmh1 jointly regulate transcription of glucose repressed genes in Saccharomyces cerevisiae. Nucleic Acids Res. 44, 5629-5645.

Liu, Q., Chang, C.E., Wooldredge, A.C., Fong, B., Kennedy, B.K., and Zhou, C. (2022). Tom70-based transcriptional regulation of mitochondrial biogenesis and aging. Elife 11

Pfanner, N., Warscheid, B., and Wiedemann, N. (2019). Mitochondrial proteins: from biogenesis to functional networks. Nat Rev Mol Cell Biol 20, 267-284.

Ruan, L., Zhou, C., Jin, E., Kucharavy, A., Zhang, Y., Wen, Z., Florens, L., and Li, R. (2017). Cytosolic proteostasis through importing of misfolded proteins into mitochondria. Nature 543, 443-446.

I prefer to have "all in one", also due to time limitation.

It would be great to be able to upload the review file as otherwise formatting and symbols get lost.

Reviewer #3 (Public Review):

In this study, Wang et al extend on their previous finding of a novel quality control pathway, the MAGIC pathway. This pathway allows misfolded cytosolic proteins to become imported into mitochondria and there they are degraded by the LON protease. Using a screen, they identify Snf1 as a player that regulates MAGIC. Snf1 inhibits mitochondrial protein import via the transcription factor Hap4 via an unknown pathway. This allows cells to adapt to metabolic changes, upon high glucose levels, misfolded proteins an become imported and degraded, while during low glucose growth conditions, import of these proteins is prevented, and instead import of mitochondrial proteins is preferred.

This is a nice and well-structured manuscript reporting on important findings about a regulatory mechanism of a quality control pathway. The findings are obtained by a combination of mostly fluorescent protein-based assays. Findings from these assays support the claims well.

While this study convincingly describes the mechanisms of a mitochondria-associated import pathway using mainly model substrates, my major concern is that the physiological relevance of this pathway remains unclear: what are endogenous substrates of the pathway, to which extend are they imported and degraded, i.e. how much does MAGIC contribute to overall misfolded protein removal (none of the experiments reports quantitative "flux" information). Lastly, it remains unclear by which mechanism Snf1 impacts on MAGIC or whether it is "only" about being outcompeted by mitochondrial precursors.

We thank Reviewer 3 for the positive and encouraging comments on our manuscript. We agree with the reviewer that identifying MAGIC endogenous substrates and understanding what percentage of them are degraded in mitochondria are very important issues to be addressed. We are indeed carrying out projects to address these questions. We also agree with Reviewer 3 that the effect of Snf1 on MAGIC may have additional mechanisms in addition to precursors competition, such as Tom6 mediated conformational changes of TOM pores. In the revised manuscript, we had added a discussion to address these comments (Page 12: line 21-28).

Reviewer #3 (Recommendations For The Authors):

- 1. In their screen, the authors utilize differences in GFP intensity as a measure for import efficiency. However, reconstitution of the GFP from GFP1-10 and GFP11 in the matrix might also be affected (folding factors, differential degradation).*

Upon Snf1 activation, the protein abundance of mitochondrial chaperones such as Hsp10, Hsp60, and Mdj1, and mitochondrial proteases such as Pim1 are not significantly changed (ref. 35). Therefore, it is unlikely that the folding and degradation capacity of mitochondrial matrix is drastically affected by Snf1 activation.

To examine the effect of Snf1 activation on spGFP reconstitution, Grx5 spGFP strain was constructed in which the endogenous mitochondrial matrix protein Grx5 was C-terminally tagged with GFP11 at its genomic locus, and GFP1-10 was targeted to mitochondria through cleavable Su9 MTS (MTS-mCherryGFP1-10) (ref. 10). Only modest reduction in Grx5 spGFP intensity was observed in LG compared to HG, and no significant difference after adjusting the GFP1-10 abundance (spGFP/mCherry ratio) (Figure 1— figure supplement 3A-D). These data suggest that any effect on spGFP reconstitution is insufficient to explain the drastic reduction of MP accumulation in mitochondria under Snf1 activation. Overall, our results

demonstrate that Snf1 activation primarily prevents mitochondrial accumulation of MPs, but not that of normal mitochondrial proteins. (Page 6: line 17-25).

We admit, however, that to fully rule out these factors, specific intra-mitochondrial folding or degradation reporter assays would be needed.

1. Scoring of protein import always takes place using fluorescence-based assays. These always require folding of the "sensors" in the matrix. An additional convincing approach that would not rely on matrix folding could be pulse chase approaches coupled to fractionation assays and immunoprecipitation.

We thank reviewer 3 for this suggestion. In our previous study, we applied two different biochemical assays: APEX proximity labeling, and mitochondrial fractionation followed by protease protection. Both confirmed the entry of misfolded proteins into mitochondria as observed by using split GFP. As we discussed in response to Reviewer 1's main point [3], the fractionation assays are not quantitative enough for the comparisons made in our study. In particular, during the over 2-hour assay, misfolded proteins continue to be degraded within mitochondria. By using proper controls, our spGFP system provides quantitative comparisons for mitochondrial accumulation of misfolded proteins in non-disturbed physiological conditions.

1. Could the pathway be reconstituted in vitro with isolated mitochondria to test for the "competition hypothesis"

This is an excellent suggestion, but setting up such a reconstituted system is a project on its own. The study documented in this manuscript already encompasses a large amount of work that we feel should be published timely.

1. Fluorescence figures are not colour blind friendly (red-green). This should be improved by changing the color scheme.

We thank reviewer 3 for pointing this out and sincerely apologize for any inconvenience. However, we are unfortunately unable to change all images within a limited time. We will adopt another color scheme in future work.

1. spGFP in human cells appears to form "spot-like" structures. What are these granules?

We indeed observed granule-like structures by spGFP labeled FUS in mitochondria, which is interesting, but we did not investigate this further because it is not a focus of this study.